

Welcome

CS 472/572
Unsupervised Machine Learning

Autoencoder 1

Week-5 -Class -1

Spring 2026

Instructor: Dr. Khondker Fariha Hossain

The Rise of Deep Learning



Artificial intelligence model detects asymptomatic Covid-19 infections through cellphone-recorded coughs

Results might provide a convenient screening tool for people who may not suspect they are infected.

Jennifer Chu | MIT News Office
October 29, 2020

Automation And Algorithms:
De-Risking Manufacturing With
Artificial Intelligence

 Sarah Goehrke Contributor @
Manufacturing
I focus on the industrialization of additive manufacturing.

TWEET THIS
The two key applications of AI in manufacturing are pricing and manufacturability feedback

HEALTH AND SCIENCE

Google's DeepMind A.I. beats doctors
in breast cancer screening trial

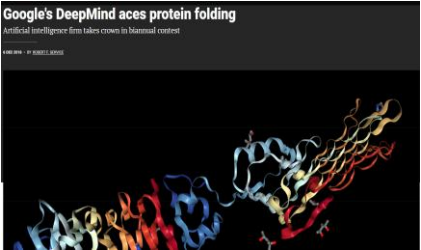
PUBLISHED THU, JAN 2 2020-8:13 AM EST | UPDATED THU, JAN 2 2020-8:13 AM EST

 David Reid
@DAVIDREID73

SHARE    



Neural networks everywhere
New chip reduces neural networks' power consumption by up to 95 percent, making them practical for battery-powered devices.



Let There Be Sight: How Deep Learning Is Helping the Blind 'See'



Predicting Cryptocurrency Prices using AI/ML

Posted By Nirav Prajapati, on April 21, 2018



'Deep Voice' Software
Can Clone Anyone's
Voice With Just 3.7
Seconds of Audio

Using snippets of voices, Baidu's 'Deep Voice' can generate new speech, accents, and tones.



The Rise of Deep Learning

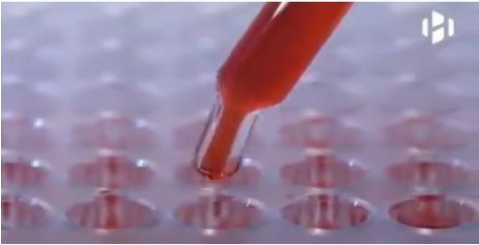
Breakthrough: Researchers trained a machine learning algorithm to detect 50 types of cancer from a drop of blood

Cancer kills 10 million people every year, of all ages. Early detection is critical to cure people

The test is based on the DNA shed by tumours and found circulating in the blood. An amazing research by Harvard's Dana-Farber Cancer Institute and the Mayo Clinic

Paper here: <https://lnkd.in/gz58B98>

#innovation #artificialintelligence #technology #tech4good



Motion Primitives-based Navigation Planning using Deep Collision Prediction

Tailored to the combined need for large-scale exploration of challenging and confined environments, despite the limited endurance of micro aerial vehicles, the proposed planner employs motion primitives to identify admissible paths

An impressive work by Kostas Alexis at the Norwegian University of Science and Technology (NTNU)

Link in the comments

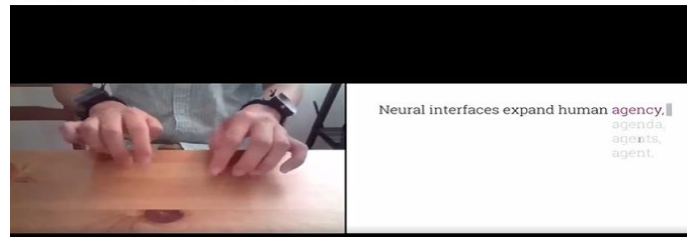


Imagine typing without a keyboard and controlling visual interfaces without a mouse

Startup CTRL-labs developed a wristband that translates musculoneural signals into machine-interpretable commands. A mind-blowing potential

CTRL-labs' vision is to get neural interfaces into commercial use by 2024. Founded in 2015, the startup was acquired by Meta last year

#innovation #technology #VR #AR



بيع أول لوحة "رسمها" الذكاء الاصطناعي

20 أكتوبر 2018 - 13:42 بتوقيت أبوظبي



Science News from research organization

Google's AI program: Building better algorithms for detecting eye disease

Date: March 13, 2018

Source: American Academy of Ophthalmology

Summary: A new study lays important groundwork for an automated system that can detect diabetic eye disease.

The First Novel Written by AI Is Here—and It's as Weird as You'd Expect It to Be

By Thomas H. Horridge - Oct 25, 2018 • 2,176

nature

Explore content ▾ About the journal ▾ Publish with us ▾ Subscribe

nature > spotlight > article

SPOTLIGHT | 30 May 2018

How artificial intelligence is changing drug discovery

Machine learning and other technologies are expected to make the hunt for new pharmaceuticals quicker, cheaper and more effective.

The Rise of Deep Learning



Machine Learning Mastery
Making Developers Awesome at Machine Learning

Search...

Get StartedBlogTopics ▾EBooksFAQAboutContact

18 Impressive Applications of Generative Adversarial Networks (GANs)

by Jason Brownlee on June 14, 2019 in Generative Adversarial Networks

Tweet

Share

in Share

Last Updated on July 12, 2019

3. Automatic Machine Translation

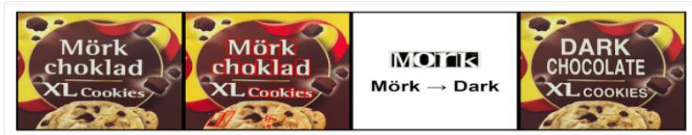
This is a task where given words, phrase or sentence in one language, automatically translate it into another language.

Automatic machine translation has been around for a long time, but deep learning is achieving top results in two specific areas:

- Automatic Translation of Text.
- Automatic Translation of Images.

Text translation can be performed without any preprocessing of the sequence, allowing the algorithm to learn the dependencies between words and their mapping to a new language. Stacked networks of large LSTM recurrent neural networks are used to perform this translation.

As you would expect, convolutional neural networks are used to identify images that have letters and where the letters are in the scene. Once identified, they can be turned into text, translated and the image recreated with the translated text. This is often called instant visual translation.



2. Automatically Adding Sounds To Silent Movies

In this task the system must synthesize sounds to match a silent video.

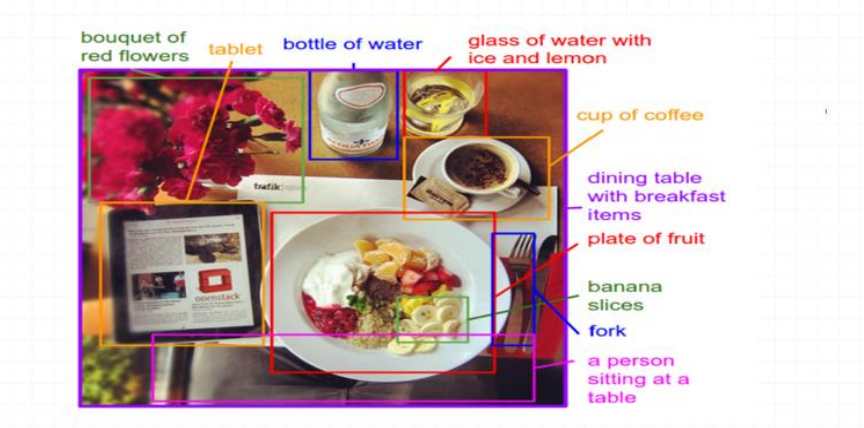
The system is trained using 1000 examples of video with sound of a drum stick striking different surfaces and creating different sounds. A deep learning model associates the video frames with a database of pre-recorded sounds in order to select a sound to play that best matches what is happening in the scene.

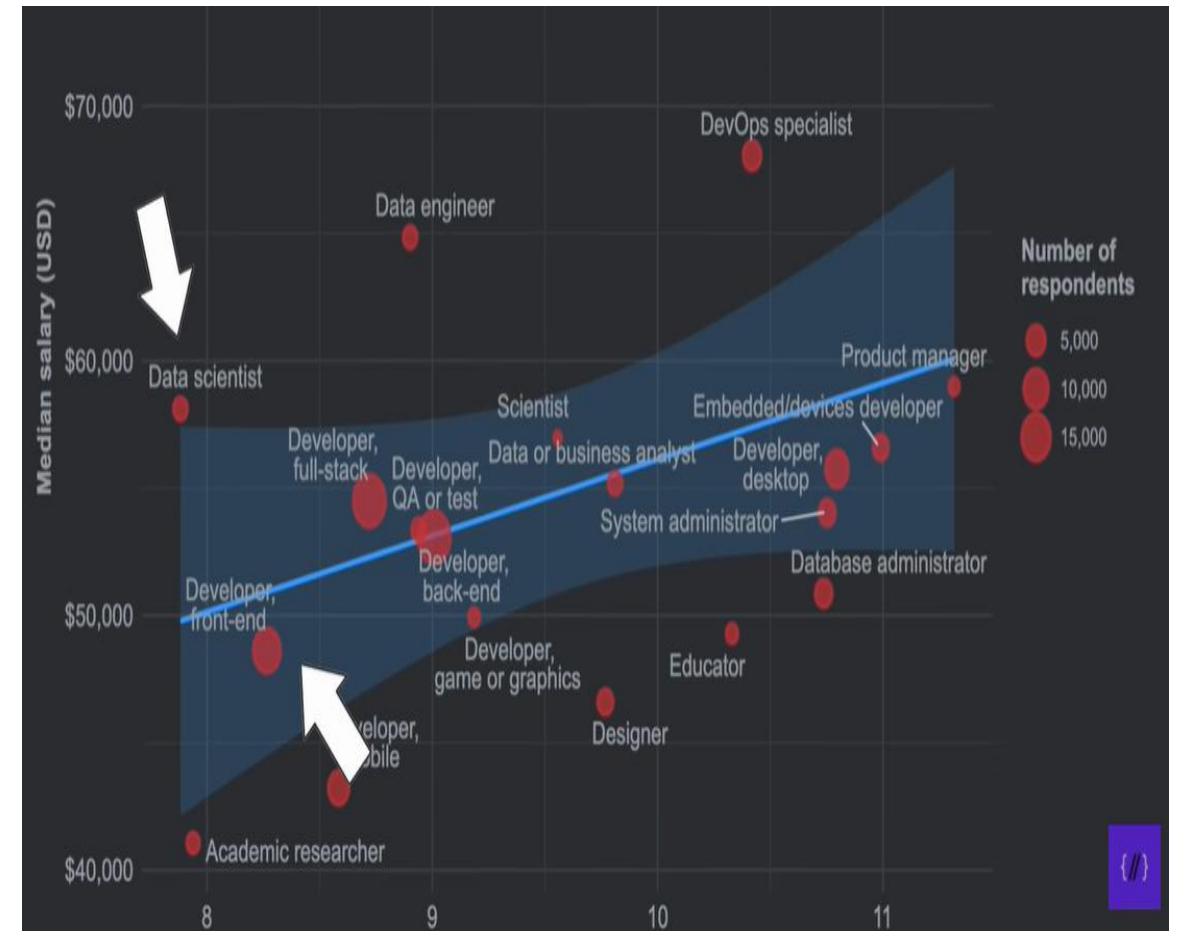
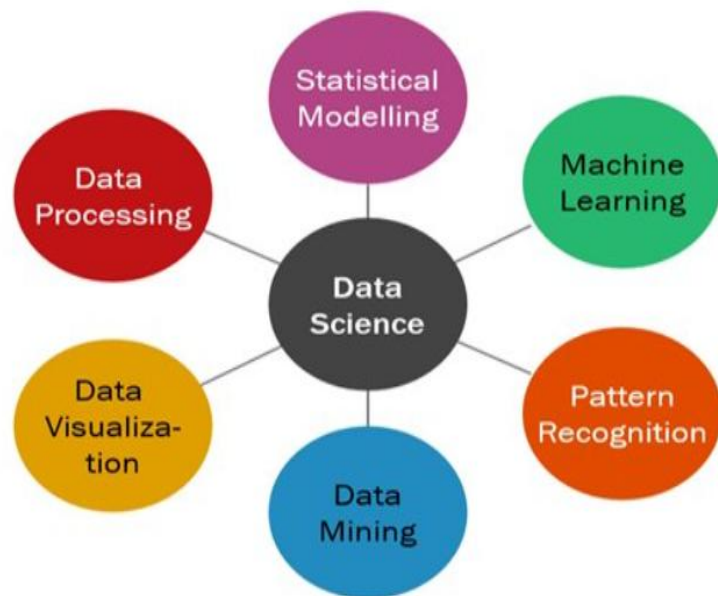
The system was then evaluated using a turing-test like setup where humans had to determine which video had the real or the fake (synthesized) sounds.

A very cool application of both convolutional neural networks and LSTM recurrent neural networks.



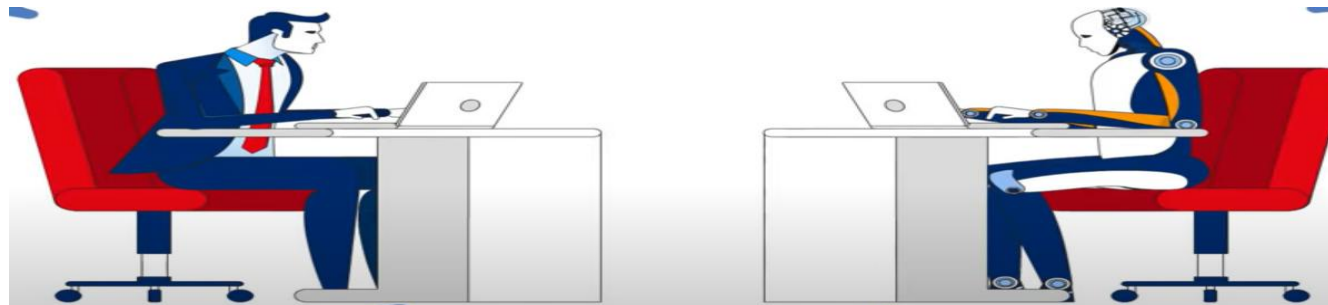
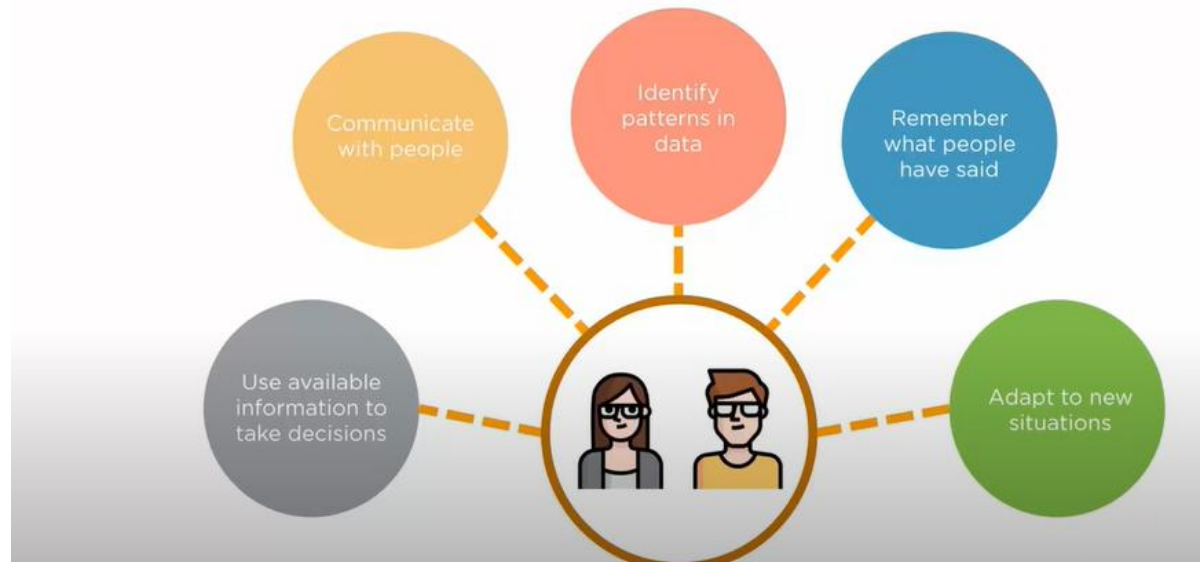
Automatic Description Generation from Images





HI vs AI

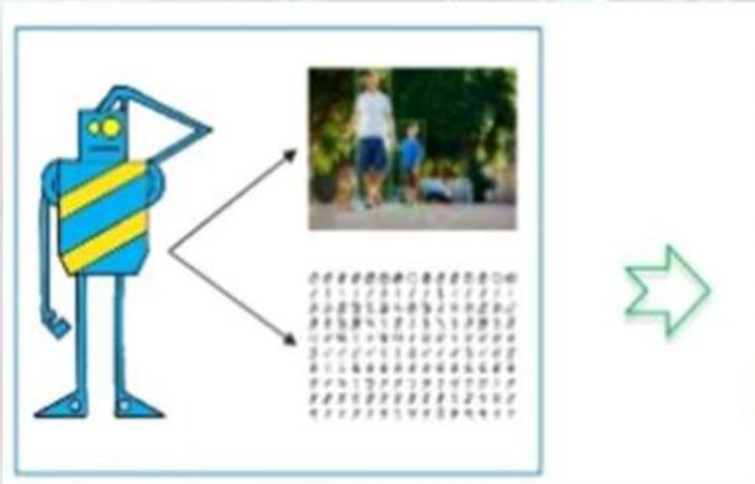
Human Intelligence



Limitation of ML

One Of The Big Challenges With Traditional Machine Learning Models Is A Process Called Feature Extraction. For Complex Problems Such As Object Recognition Or Handwriting Recognition, This Is A Huge Challenge.

Deep Learning To The Rescue



Deep Learning Models Are Capable Of Focus On The Right Features By Themselves, Requiring Little Guidance From The Programmer. These Models Can Also Partially Solve The Dimensionality Problem.



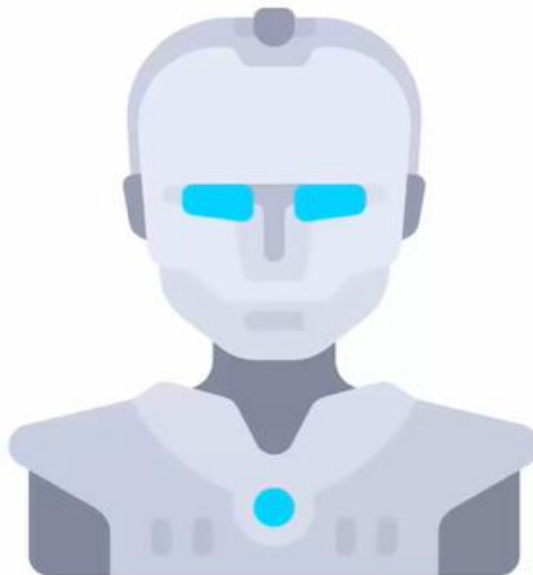
The Idea Behind Deep Learning Is To Build Learning Algorithms That Mimic Brain

Artificial Intelligence

AI develops computer systems that can accomplish tasks that require human intelligence

Interacts with humans using their natural language

Provides more accurate results



Learn from their mistakes and adapt to new environments

Learns from the data, and automates repetitive learning

simplilearn

simplilearn. All rights reserved.

What is Deep Learning?

ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



MACHINE LEARNING

Ability to learn without explicitly being programmed



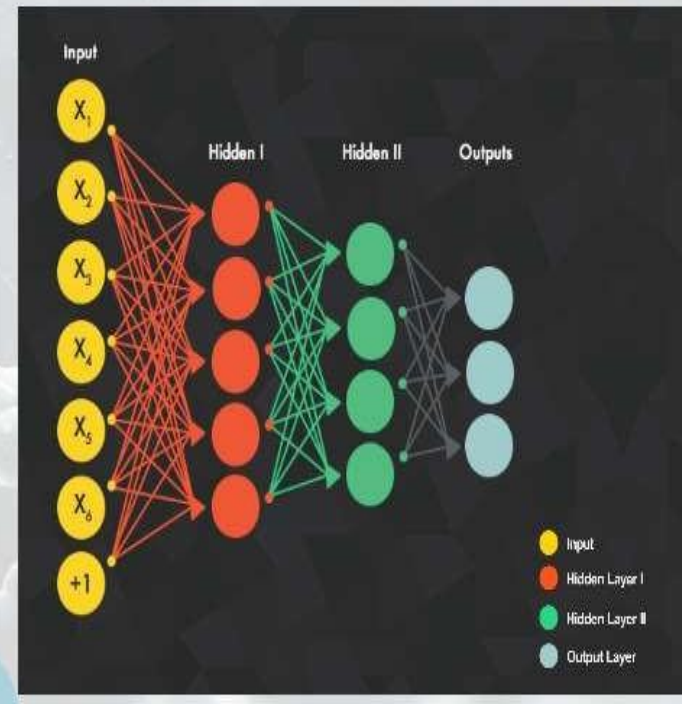
DEEP LEARNING

Extract patterns from data using neural networks



What is Deep Learning(DL)

- A class of machine learning techniques that exploit many layer of non linear information processing for supervised or unsupervised feature extraction and transformation and for pattern analysis and classification (Deng & Yu, 2014)
- It typically uses deep artificial neural networks.



Adapted from: www.edureka.co

What is Deep Learning(DL)

- A sub-field of machine learning that is based on learning several levels of representations corresponding to a hierarchy of features or factors or concepts, where higher-level concepts are defined from lower-level ones, and the same lower level concepts can help to define many higher-level concepts.

(LeCun, Bengio, & Hinton, 2015).

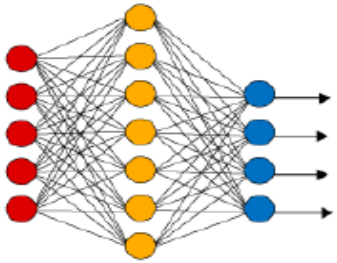
Deep Learning = Machine Learning + Representation Learning

- *Representation learning is a set of methods that allows a machine to be fed with raw data and to automatically discover the representations needed for detection or classification.*

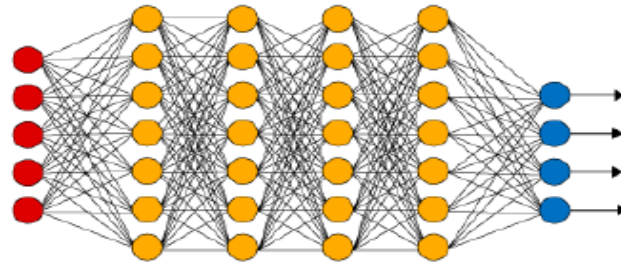
Adapted from: www.edureka.co

What is Deep Learning(DL)

Simple Neural Network



Deep Learning Neural Network



● Input Layer ● Hidden Layer ● Output Layer



Part or a powerful class of the machine learning field of learning representations of data. Exceptional effective at learning patterns.



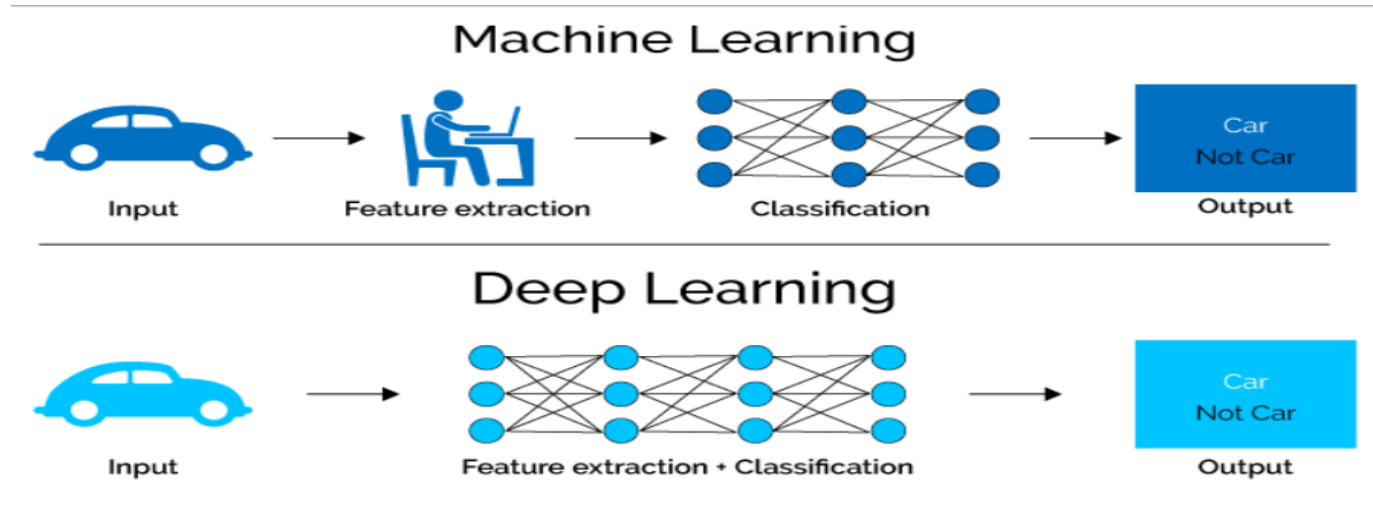
Utilize learning algorithms that derive meaning out of data by using hierarchy of multiple layers that mimic the neural networks of our brain.



If you provide the system tons of information , it begins to understand it and respond in useful ways.

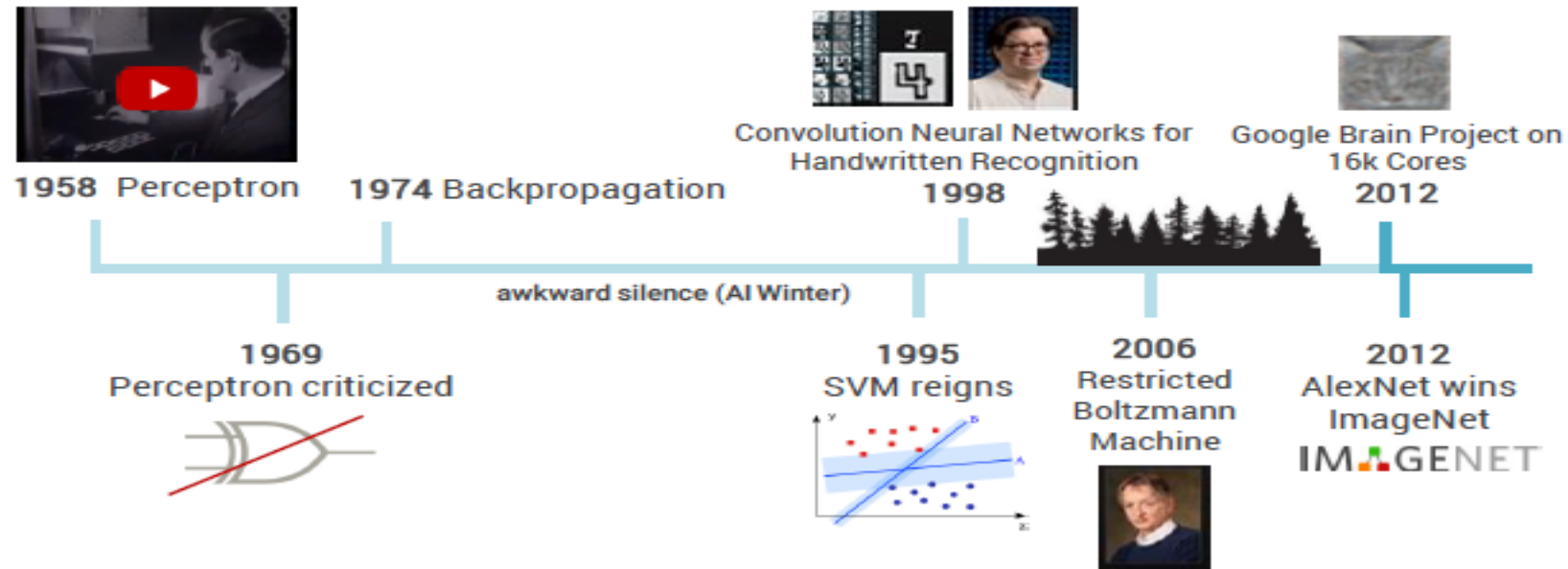
Stacked “Neural Network”. Is usually indicates “Deep Neural Network”.

What is Deep Learning(DL)



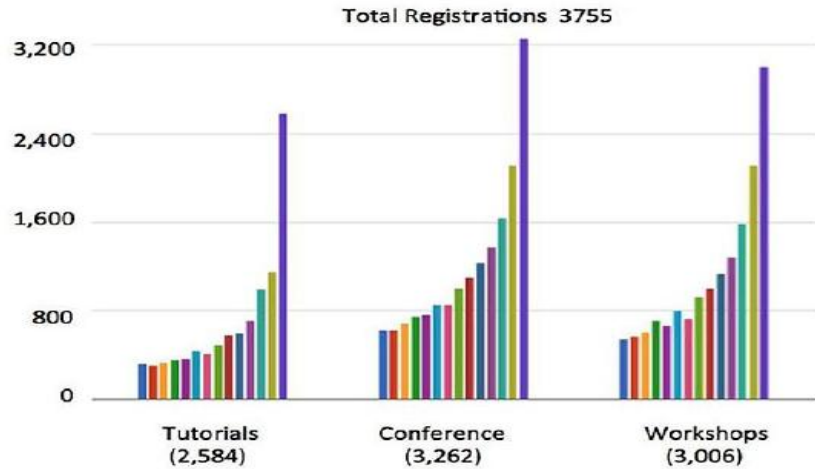
- Collection of simple, trainable mathematical functions.
- Learning methods which have deep architecture.
- It often allows end-to-end learning.
- It automatically finds intermediate representation. Thus, it can be regarded as a representation learning.

Brief History

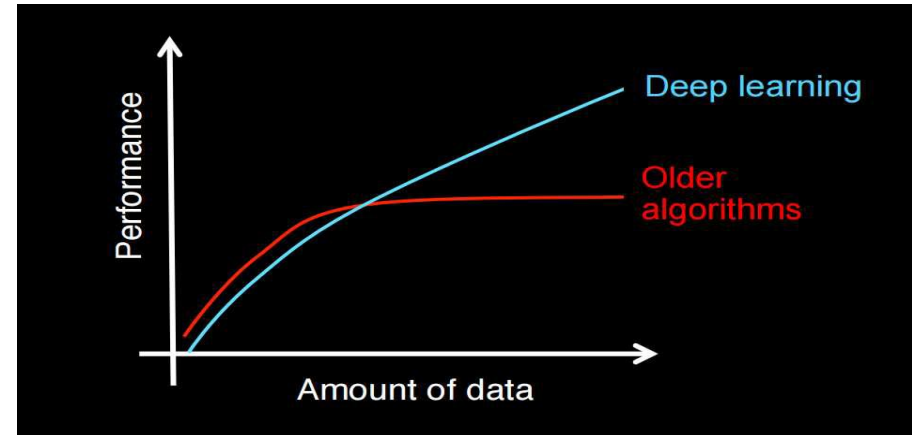


- 2012 was the first year that neural nets grew to prominence as Alex Krizhevsky used them to win that year's ImageNet competition (basically, the annual Olympics of computer vision), dropping the classification error record from 26% to 15%, an astounding improvement at the time.

Motivations

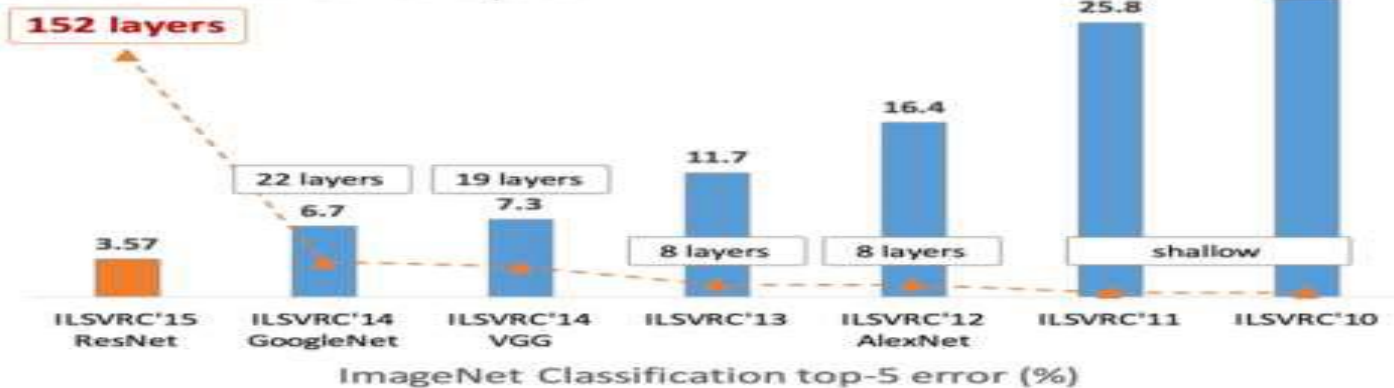


NIPS(Computational Neuroscience Conference) Growth



Amount of Data vs Performance

Revolution of Depth



Why is Deep Learning(DL) Happening now

- Actually it is not - first papers published in '970s
- Third resurgence of neura networks
 - Research breakthroughs

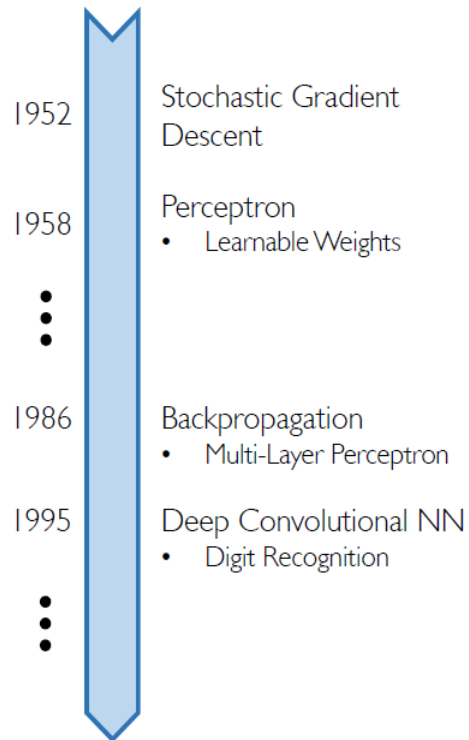
	Problem	Solution
Vanishing Gradient	Sigmoid type activation functions easily saturate. Gradients are small. in deeper layers updates become almost zero.	Earlier: layer-by-layer pretraining Recently: non-saturating Activation functions
Gradient Descent	First order methods (e.g. SGD) are easily stuck. Second order methods are feasible on larger data.	Adaptive training: adagrad, adam, adadelta, RMSProp Nestrov momentum
Regularisation	Networks easily overfit (even with l2 regularization)	Dropout

- Increase in computational power
- GP GPUs

Adapted from: www.edureka.co

Why Now?

Neural Networks date back decades, so why the resurgence?



1. Big Data

- Larger Datasets
- Easier Collection & Storage

IMAGENET



2. Hardware

- Graphics Processing Units (GPUs)
- Massively Parallelizable



3. Software

- Improved Techniques
- New Models
- Toolboxes

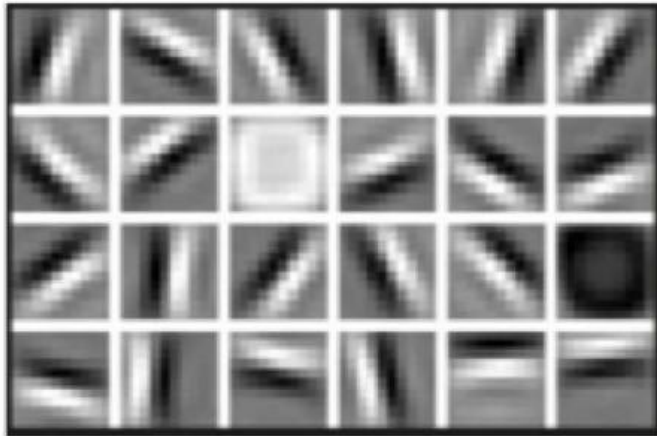


Why Deep Learning?

Hand engineered features are time consuming, brittle and not scalable in practice

Can we learn the **underlying features** directly from data?

Low Level Features



Lines & Edges

Mid Level Features



Eyes & Nose & Ears

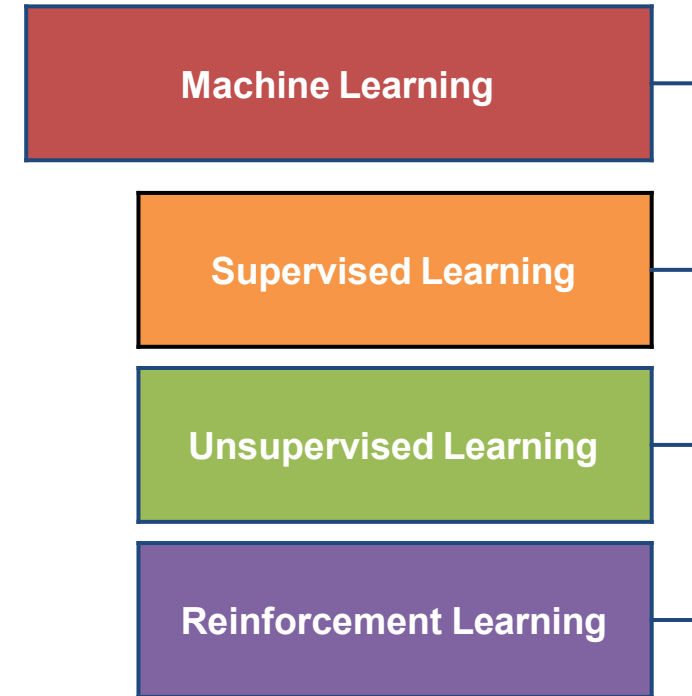
High Level Features



Facial Structure

Types of ML Algorithms

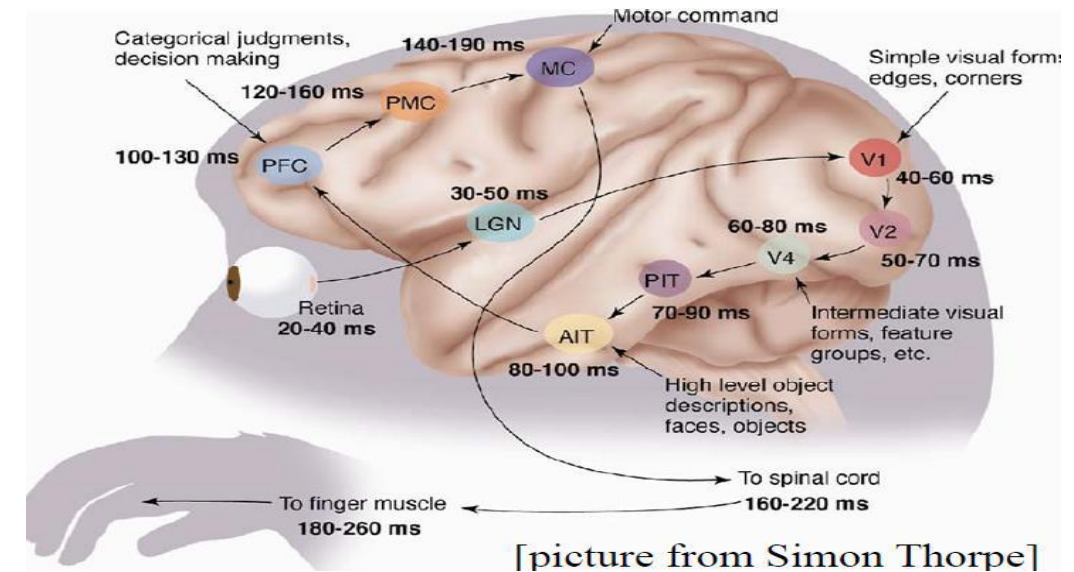
- Supervised Learning
 - trained with labeled data; including regression and classification problems
- Unsupervised Learning
 - trained with unlabeled data; clustering and association rule learning problems.
- Reinforcement Learning
 - no training data; stochastic Markov decision process; robotics and self-driving cars.



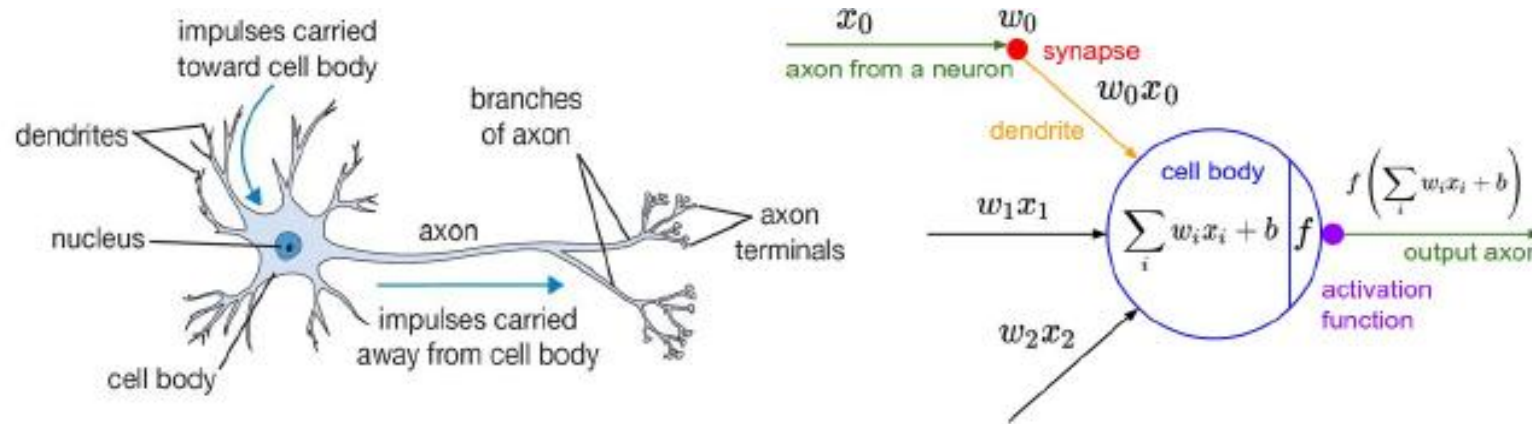
Deep Learning Basics :Neuron

Commonalities with real brains:

- Each neuron is connected to a small subset of other neurons.
- Based on what it sees, it decides what it wants to say.
- Neurons learn to cooperate to accomplish the task.
- The ventral pathway in the visual cortex has multiple stages
- There exist a lot of intermediate representations



Deep Learning Basics :Neuron



An artificial neuron contains a nonlinear activation function and has several incoming and outgoing weighted connections.

Neurons are trained filters and detect specific features or patterns (e.g. edge, nose) by receiving weighted input, transforming it with the activation function and passing it to the outgoing connections

Unique Characteristics of DL

- Automatic feature extraction
- Makes use of High-dimensional data for training
- Defines higher level features from lower level features obtained from the previous layer
- It is a representation learning method
- Can solve real-world or unconstrained problems such as NLP, image recognition etc

Adapted from: www.edureka.co



Easy for
Humans



What about Machines?

The Problem Space : inputs & outputs



What We See

```
08 02 22 97 38 15 00 40 00 75 04 05 07 78 52 12 50 77 91 08
49 49 99 40 17 81 18 57 60 87 17 40 98 43 69 48 04 56 62 00
81 49 31 73 55 79 14 29 93 71 40 67 53 88 30 03 49 13 36 65
52 70 95 23 04 60 11 42 49 24 68 56 01 32 56 71 37 02 36 91
22 31 16 71 51 67 63 89 41 92 36 54 22 40 40 28 66 33 13 80
24 47 32 60 99 03 45 02 44 75 33 53 78 36 84 20 35 17 12 50
32 98 81 28 64 23 47 10 26 38 40 47 59 54 70 66 18 38 64 70
67 24 20 68 02 62 12 20 95 63 94 39 63 08 40 91 66 49 94 21
24 55 58 05 66 73 99 26 97 17 78 78 96 83 14 88 34 89 63 72
21 36 23 09 75 00 76 44 20 45 35 14 00 61 93 97 34 31 93 95
78 17 53 28 22 75 31 67 15 94 03 80 04 62 16 14 09 53 56 92
16 39 05 42 96 35 31 47 55 58 88 24 00 17 54 24 36 29 85 57
86 56 00 48 35 71 89 07 05 44 44 37 44 60 21 58 51 54 17 58
19 80 81 68 05 94 47 69 28 73 92 13 86 52 17 77 04 89 55 40
04 52 08 83 97 35 99 14 07 97 57 32 16 26 26 79 33 27 92 66
88 36 68 87 57 62 20 72 03 46 33 67 46 55 12 32 63 93 53 69
04 42 16 73 38 25 39 11 24 94 72 18 08 46 29 32 40 62 76 36
20 69 36 41 72 30 23 88 34 62 99 69 82 67 59 85 74 04 36 16
20 73 35 29 78 31 90 01 74 31 49 71 48 86 81 16 23 57 05 54
01 70 54 71 83 51 54 69 16 92 33 48 61 43 52 01 89 19 47 48
```

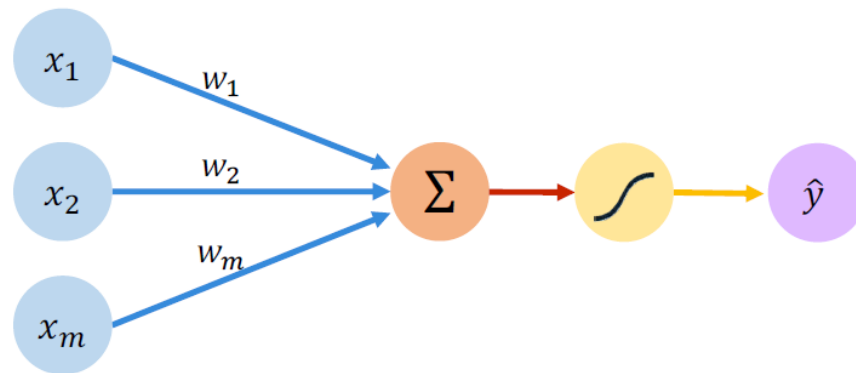
What Computers See

- Depending on the resolution and size of the image, it will see a 32 x 32 x 3 array of numbers
- These numbers, while meaningless to us when we perform image classification, are the only inputs available to the computer.
- The idea is that you give the computer this array of numbers and it will output numbers that describe the probability of the image being a certain class (.80 for cat, .15 for dog, .05 for bird, etc).

The Problem Space :What We Want the Computer to Do

- To be able to differentiate between all the images it's given and figure out the unique features that make a dog a dog or that make a cat a cat.
- This is the process that goes on in our minds subconsciously as well.
- When we look at a picture of a dog, we can classify it as such if the picture has identifiable features such as paws or 4 legs.
- In a similar way, the computer is able perform image classification by looking for low level features such as edges and curves, and then building up to more abstract concepts through a series of convolutional layers.

The Perceptron: Forward Propagation



Output

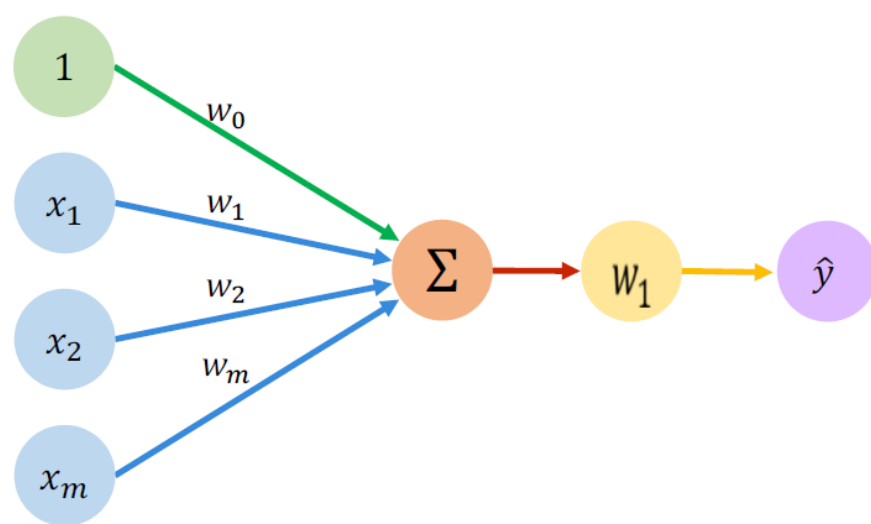
Linear combination of inputs

$$\hat{y} = g \left(\sum_{i=1}^m x_i w_i \right)$$

Non-linear activation function

Inputs Weights Sum Non-Linearity Output

The Perceptron: Forward Propagation



Inputs Weights Sum Non-Linearity Output

Linear combination of inputs

Output

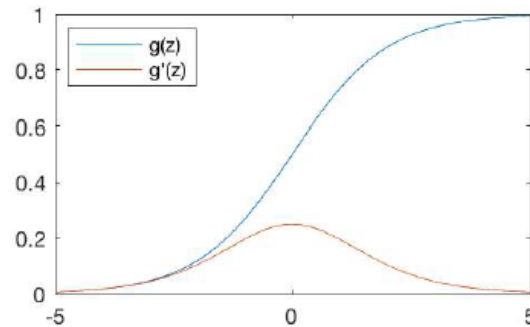
$$\hat{y} = g \left(w_0 + \sum_{i=1}^m x_i w_i \right)$$

Non-linear activation function

Bias

Common Activation Functions

Sigmoid Function



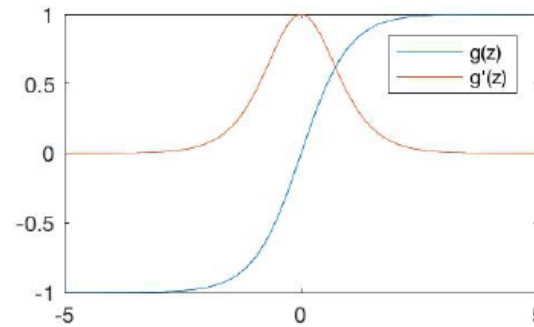
$$g(z) = \frac{1}{1 + e^{-z}}$$

$$g'(z) = g(z)(1 - g(z))$$



tf.nn.sigmoid(z)

Hyperbolic Tangent



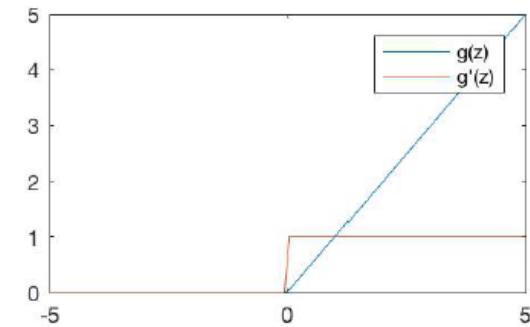
$$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$g'(z) = 1 - g(z)^2$$



tf.nn.tanh(z)

Rectified Linear Unit (ReLU)



$$g(z) = \max(0, z)$$

$$g'(z) = \begin{cases} 1, & z > 0 \\ 0, & \text{otherwise} \end{cases}$$

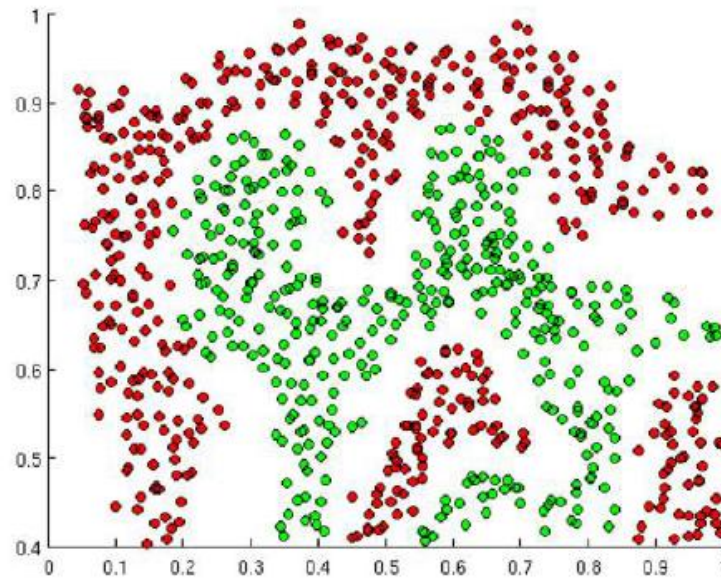


tf.nn.relu(z)

NOTE: All activation functions are non-linear

Importance of Activation Functions

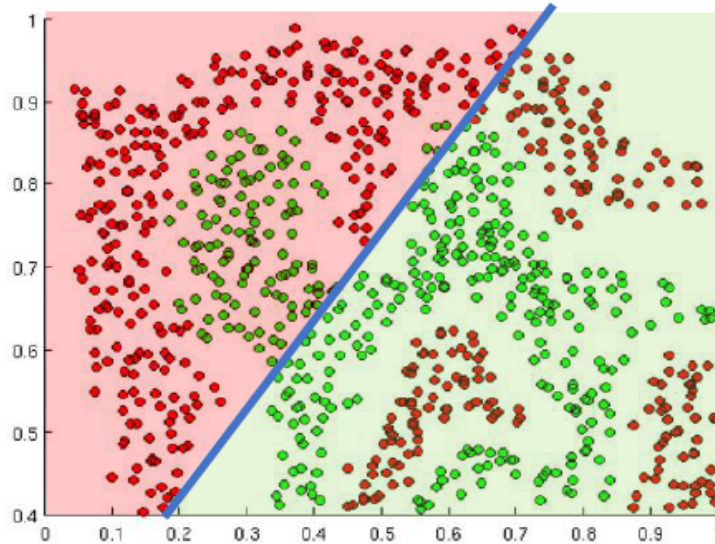
The purpose of activation functions is to *introduce non-linearities* into the network



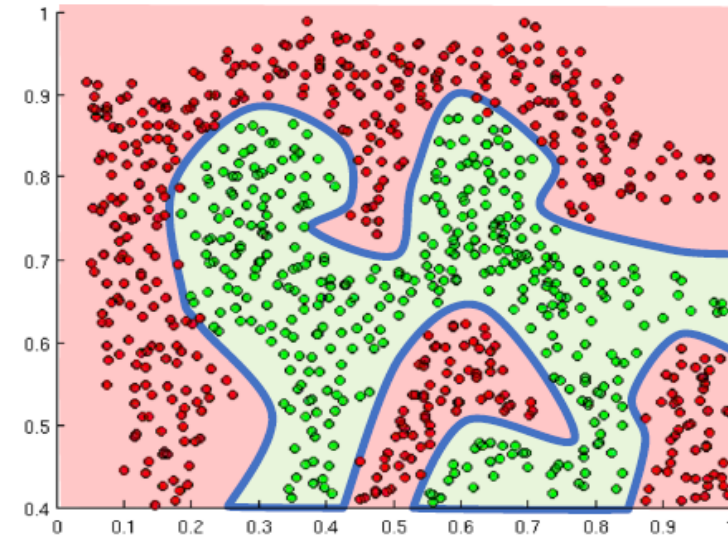
What if we wanted to build a Neural Network to distinguish green vs red points?

Importance of Activation Functions

*The purpose of activation functions is to **introduce non-linearities** into the network*

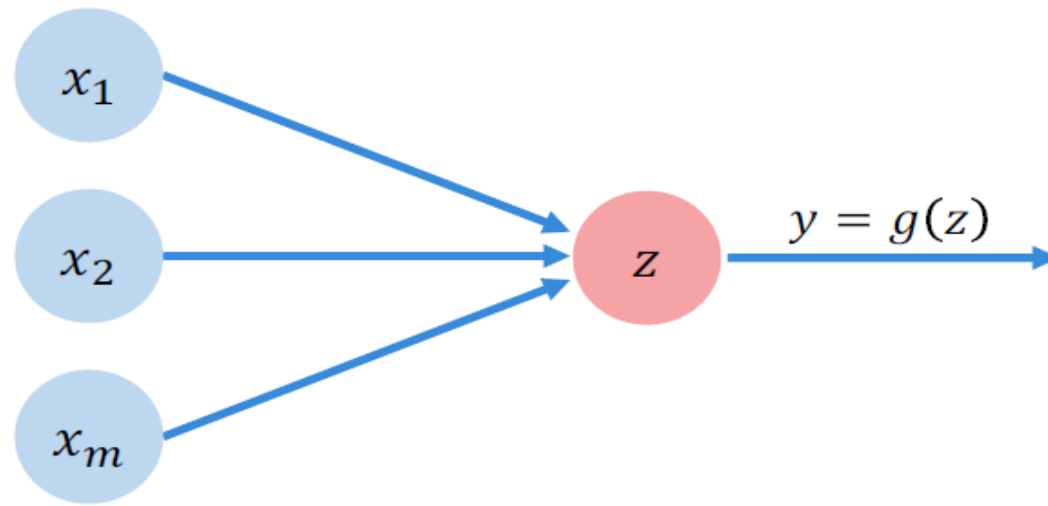


Linear Activation functions produce linear decisions no matter the network size



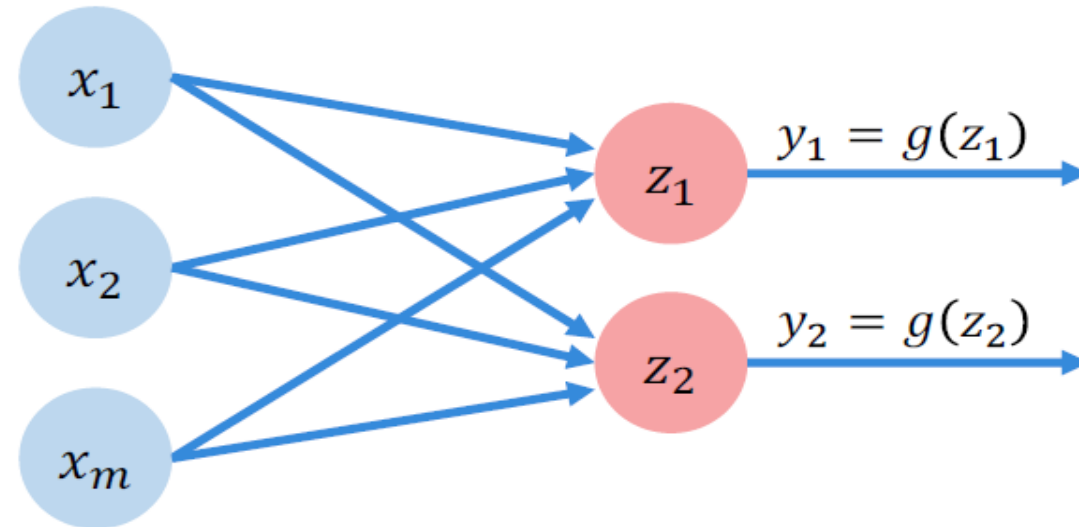
Non-linearities allow us to approximate arbitrarily complex functions

The Perceptron: Simplified



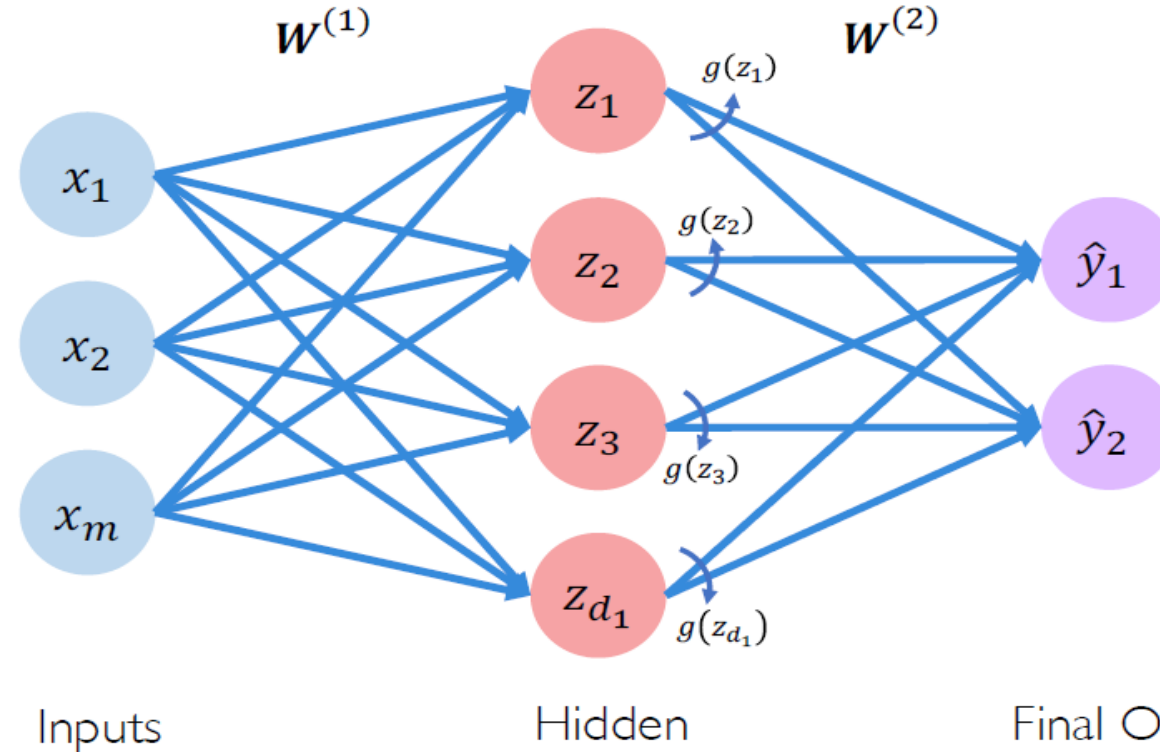
$$z = w_0 + \sum_{j=1}^m x_j w_j$$

Multi Output Perceptron



$$z_i = w_{0,i} + \sum_{j=1}^m x_j w_{j,i}$$

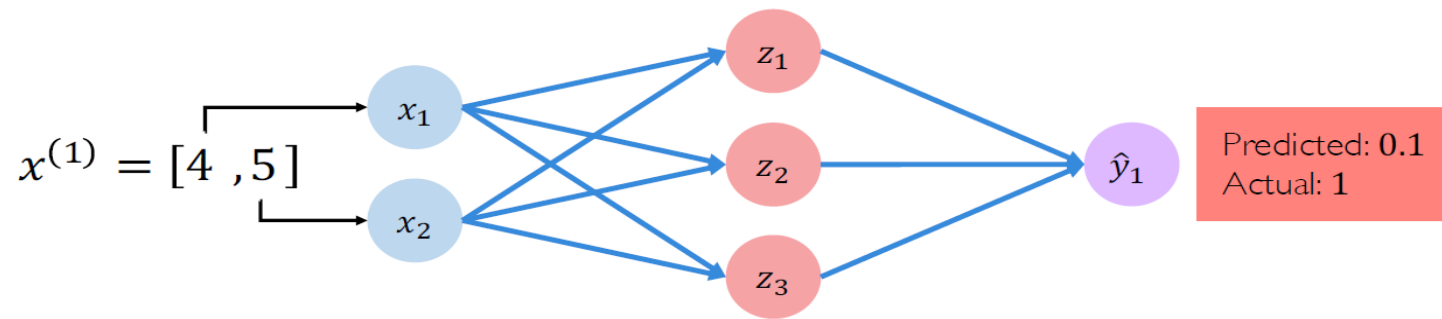
Single Layer Neural Network



$$z_i = w_{0,i}^{(1)} + \sum_{j=1}^m x_j w_{j,i}^{(1)} \quad \hat{y}_i = g \left(w_{0,i}^{(2)} + \sum_{j=1}^{d_1} z_j w_{j,i}^{(2)} \right)$$

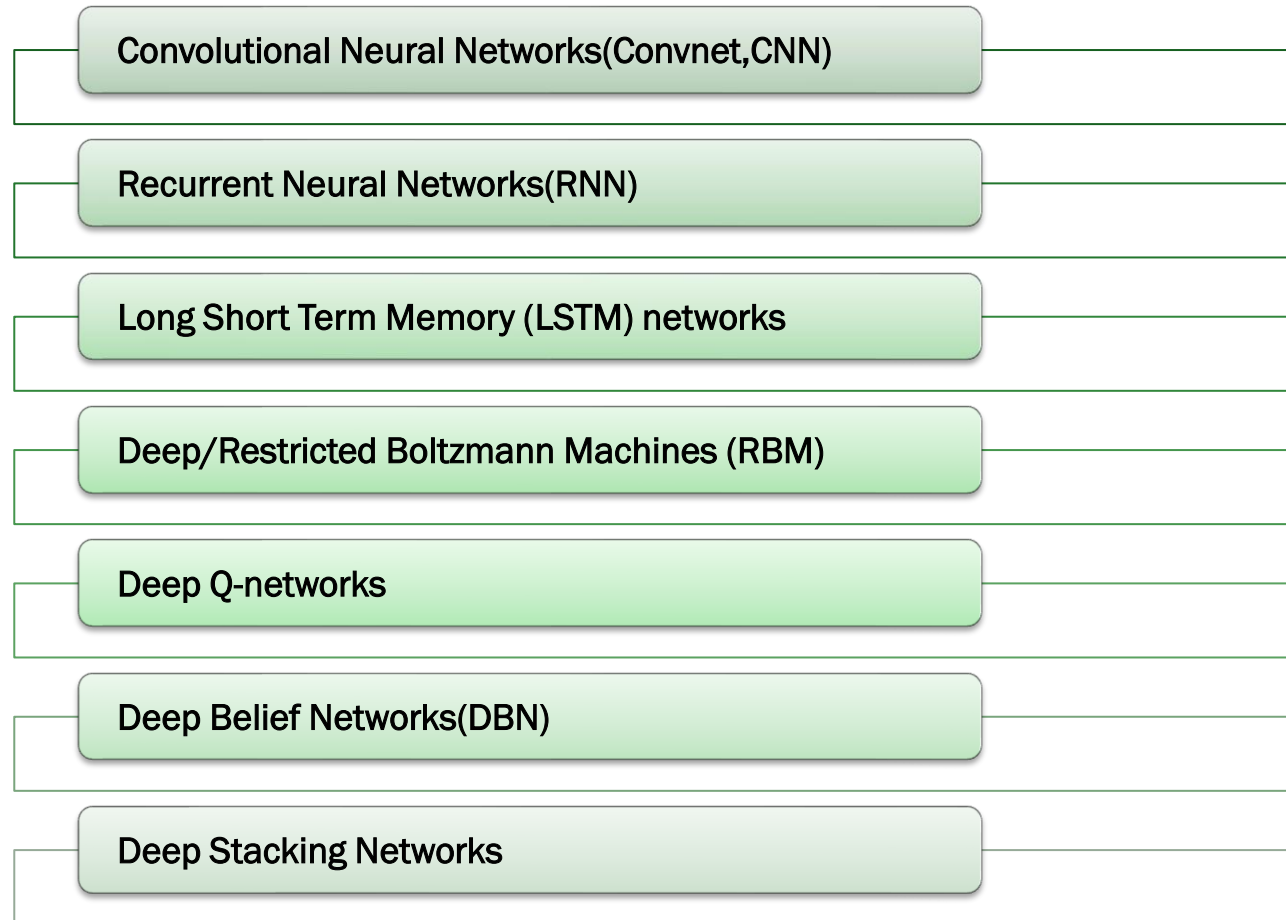
Quantifying Loss

The **loss** of our network measures the cost incurred from incorrect predictions



$$\mathcal{L}(\underbrace{f(x^{(i)}; \mathbf{W})}_{\text{Predicted}}, \underbrace{y^{(i)}}_{\text{Actual}})$$

Types of Networks used for Deep Learning



Convolution Neural Networks(CNN):Basic Idea

- This idea was expanded upon by a fascinating experiment by Hubel and Wiesel in 1962 where they showed that some individual neuronal cells in the brain responded (or fired) only in the presence of edges of a certain orientation.
- For example, some neurons fired when exposed to vertical edges and some when shown horizontal or diagonal edges. Hubel and Wiesel found out that all of these neurons were organized in a columnar architecture and that together, they were able to produce visual perception.
- This idea of specialized components inside of a system having specific tasks (the neuronal cells in the visual cortex looking for specific characteristics) is one that machines use as well, and is the basis behind CNNs.

Inputs and Outputs

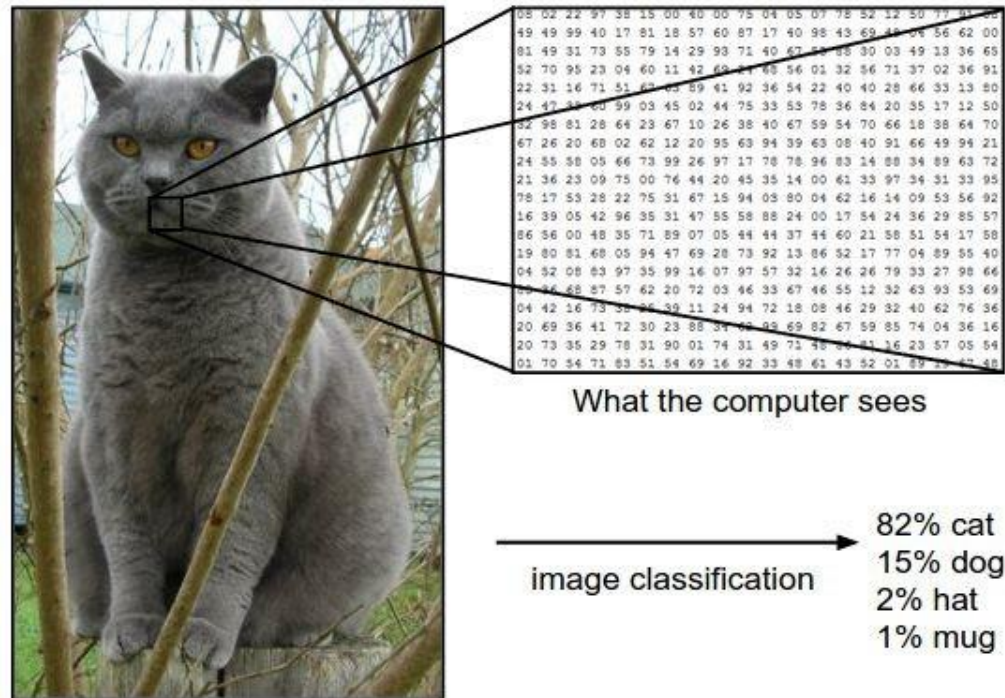
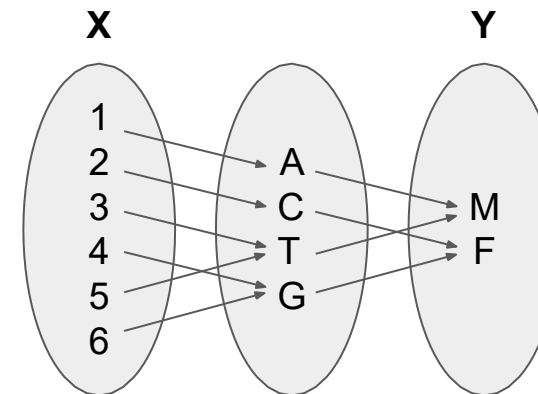
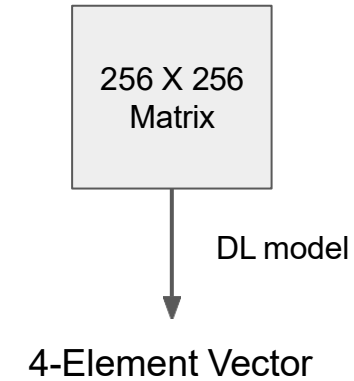
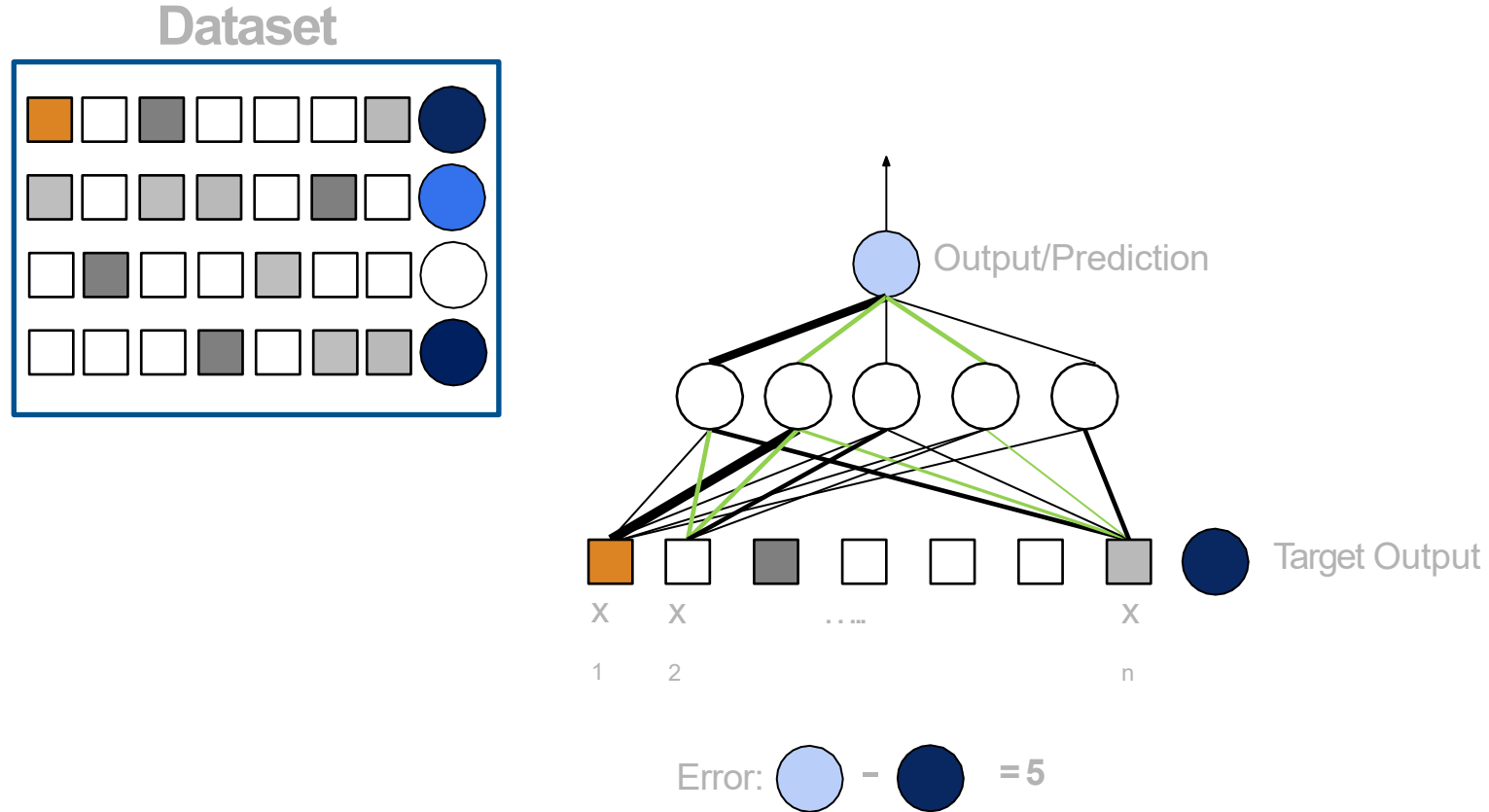


Image from the [Stanford CS231 Course](#)



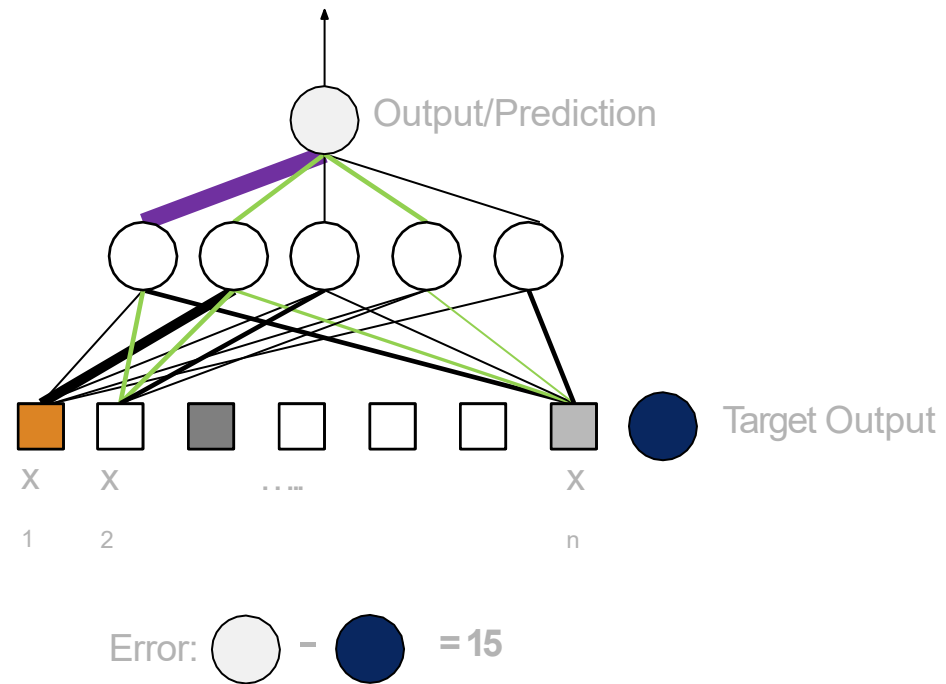
With deep learning, we are searching for a **surjective** (or **onto**) function f from a set X to a set Y .

Learning Principle



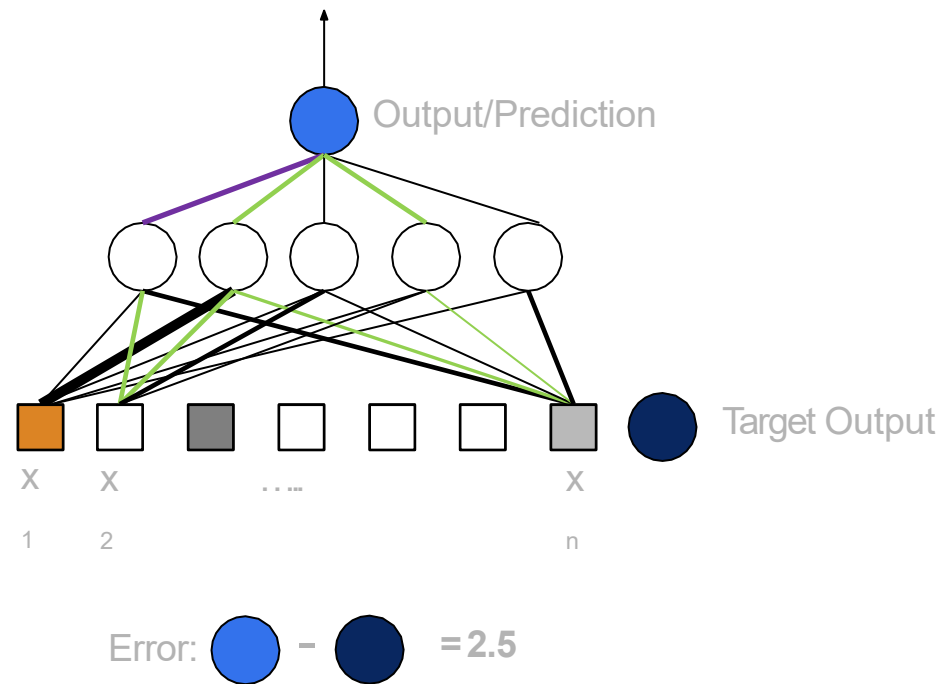
(Image Credit: NVIDIA Deep Learning Institute)

Learning Principle



(Image Credit: NVIDIA Deep Learning Institute)

Learning Principle

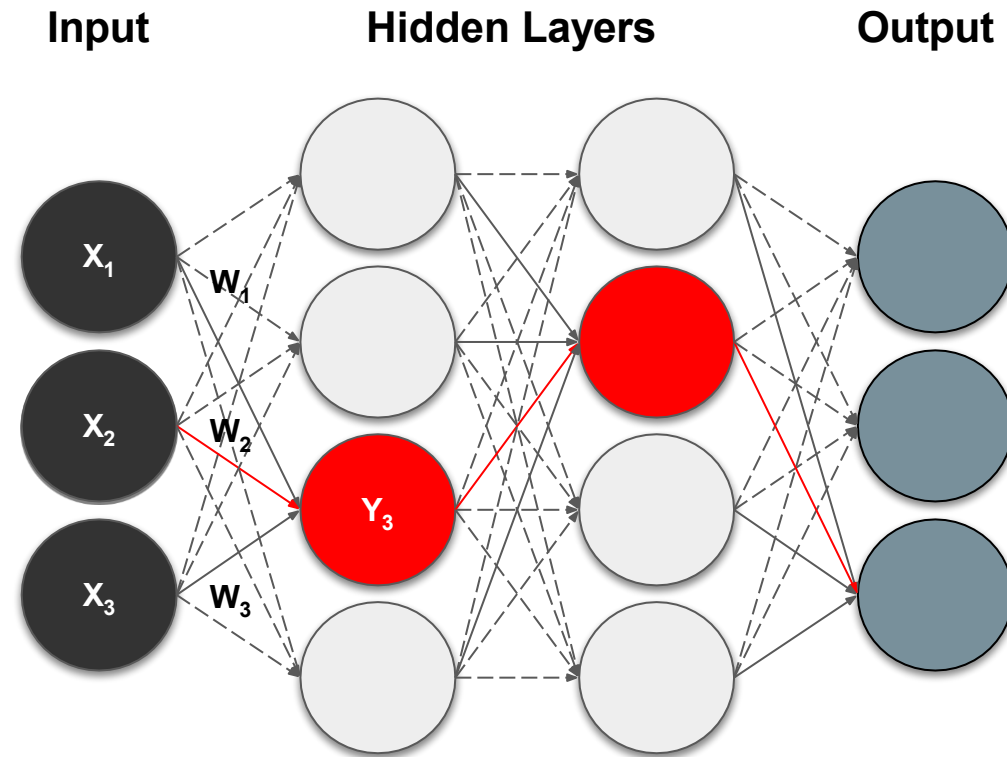


Supervised Deep Learning with Neural Networks

From one layer to the next

$$Y_j = f\left(\sum_i W_i X_i + b_i\right)$$

f is the activation function,
 W_i is the weight, and b_i is
the bias.



Training - Minimizing the Loss

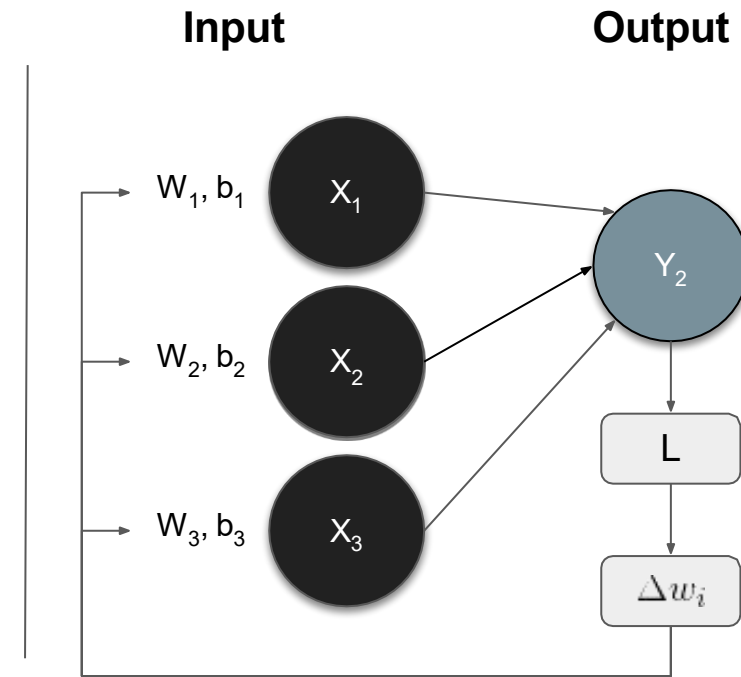
The loss function with regard to weights and biases can be defined as

$$L(\mathbf{w}, \mathbf{b}) = \frac{1}{2} \sum_i (\mathbf{Y}(\mathbf{X}, \mathbf{w}, \mathbf{b}) - \mathbf{Y}'(\mathbf{X}, \mathbf{w}, \mathbf{b}))^2$$

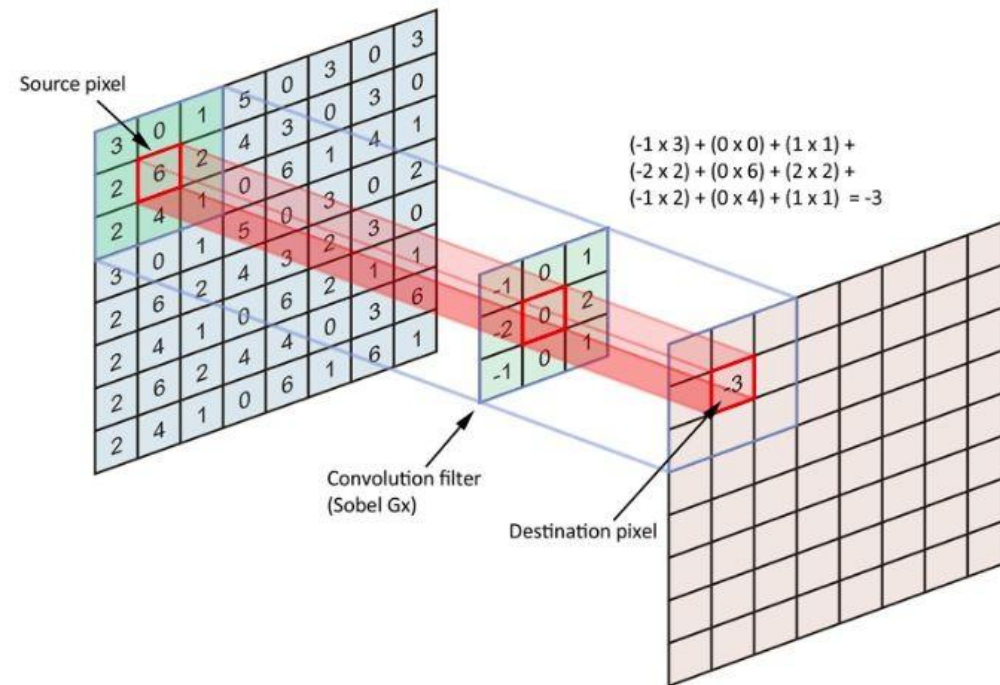
The weight update is computed by moving a step to the opposite direction of the cost gradient.

$$\Delta w_i = -\alpha \frac{\partial L}{\partial w_i}$$

Iterate until L stops decreasing.



Convolution in 2D



(Image Credit: [Applied Deep Learning | Arden Dertat](#))

Convolution Kernel

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Input

1	0	1
0	1	0
1	0	1

Filter / Kernel

1x1	1x0	1x1	0	0
0x0	1x1	1x0	1	0
0x1	0x0	1x1	1	1
0	0	1	1	0
0	1	1	0	0

4		

(Image Credit: [Applied Deep Learning | Arden Dertat](#))

Convolution on Image

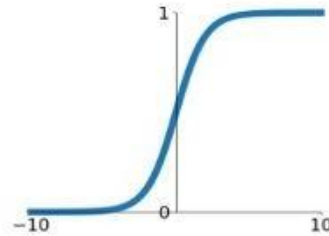


Image Credit: [Deep Learning Methods for Vision | CVPR 2012 Tutorial](#)

Activation

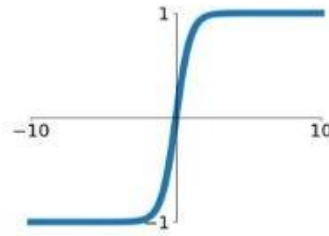
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



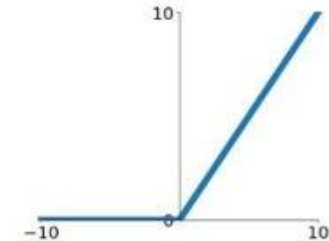
tanh

$$\tanh(x)$$



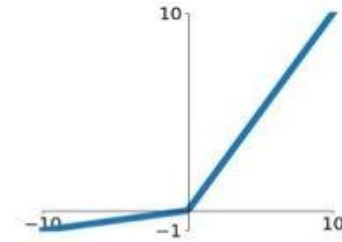
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

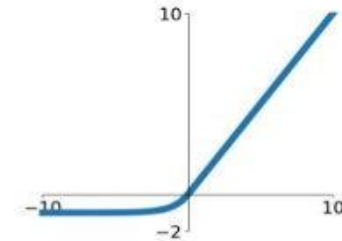


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



Introducing Non-Linearity (ReLU)

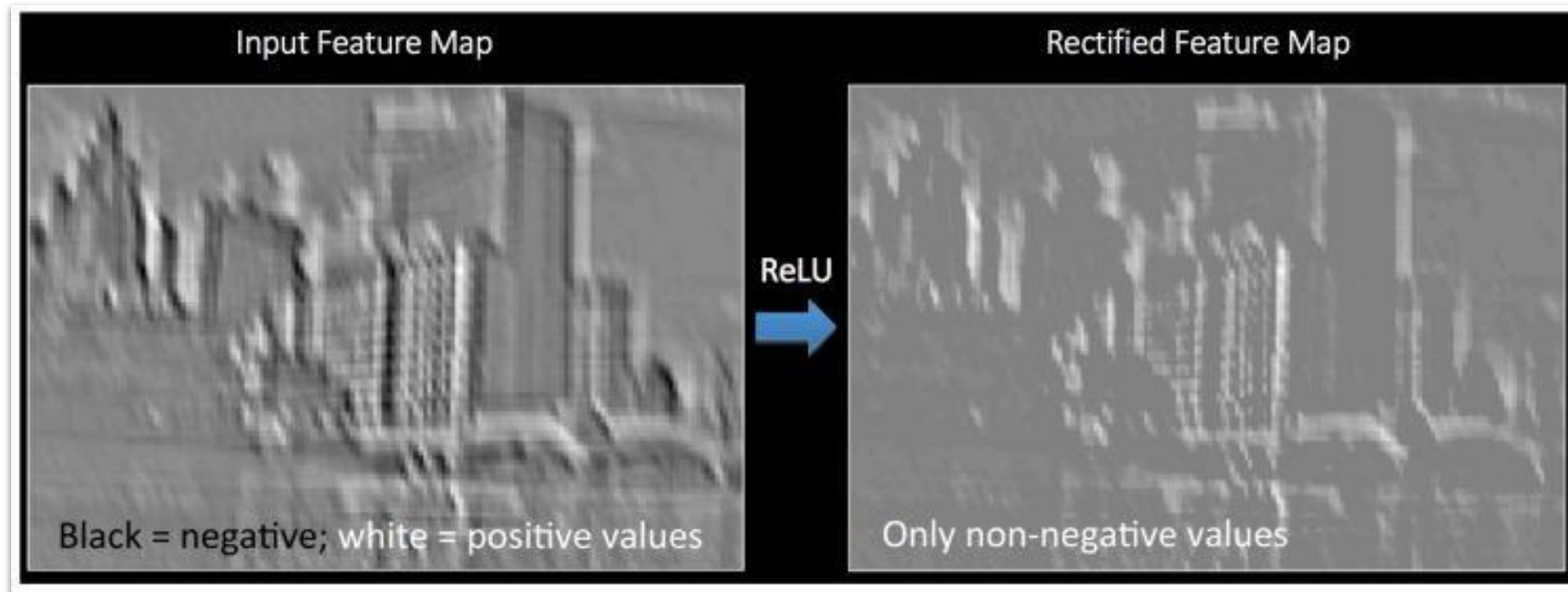
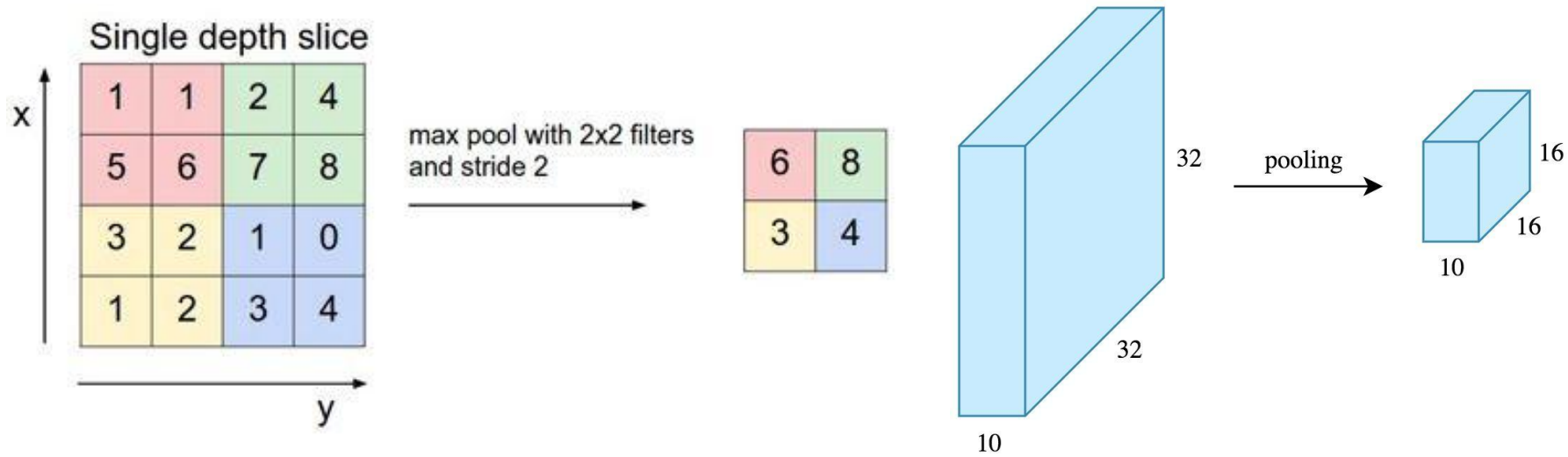


Image Credit: [Deep Learning Methods for Vision | CVPR 2012 Tutorial](#)

Max Pooling



(Image Credit: [Applied Deep Learning | Arden Dertat](#))

Pooling - Max-Pooling and Sum-Pooling

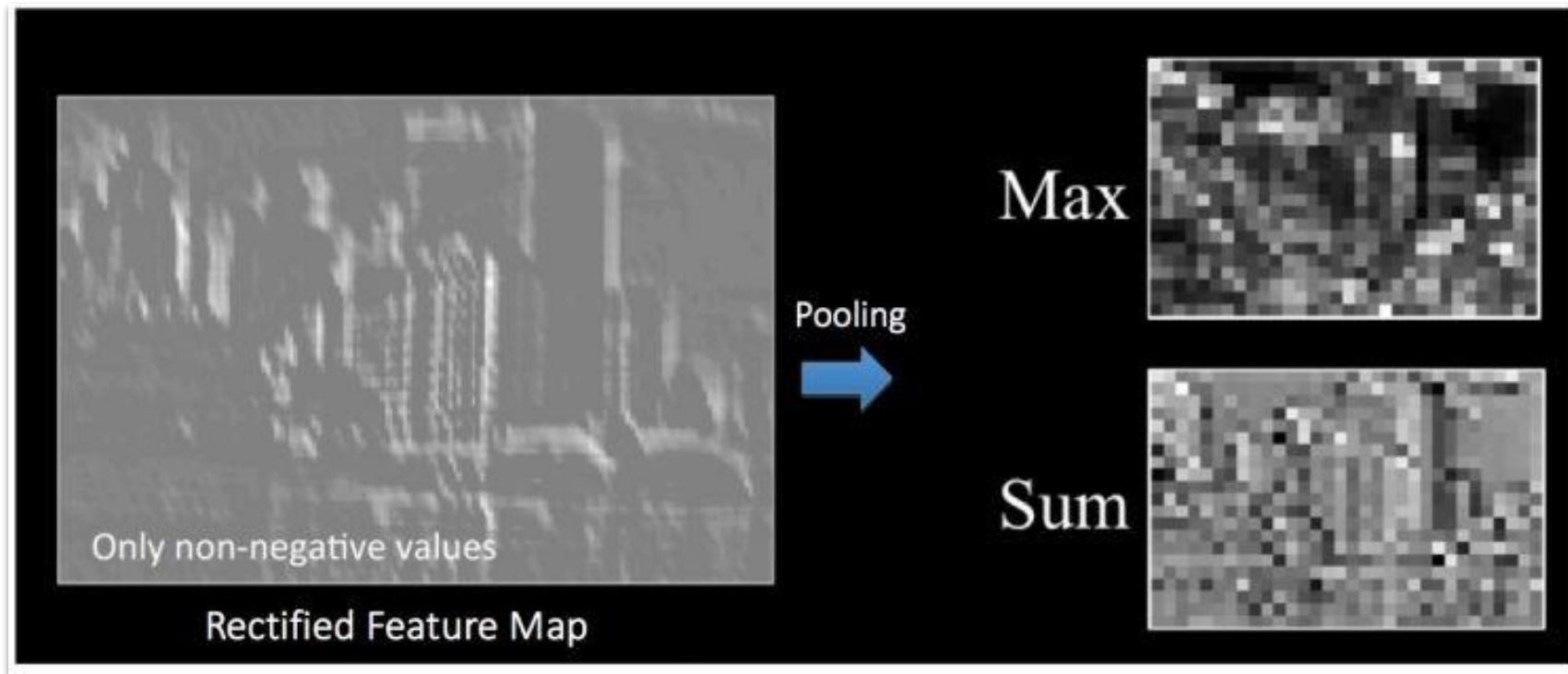
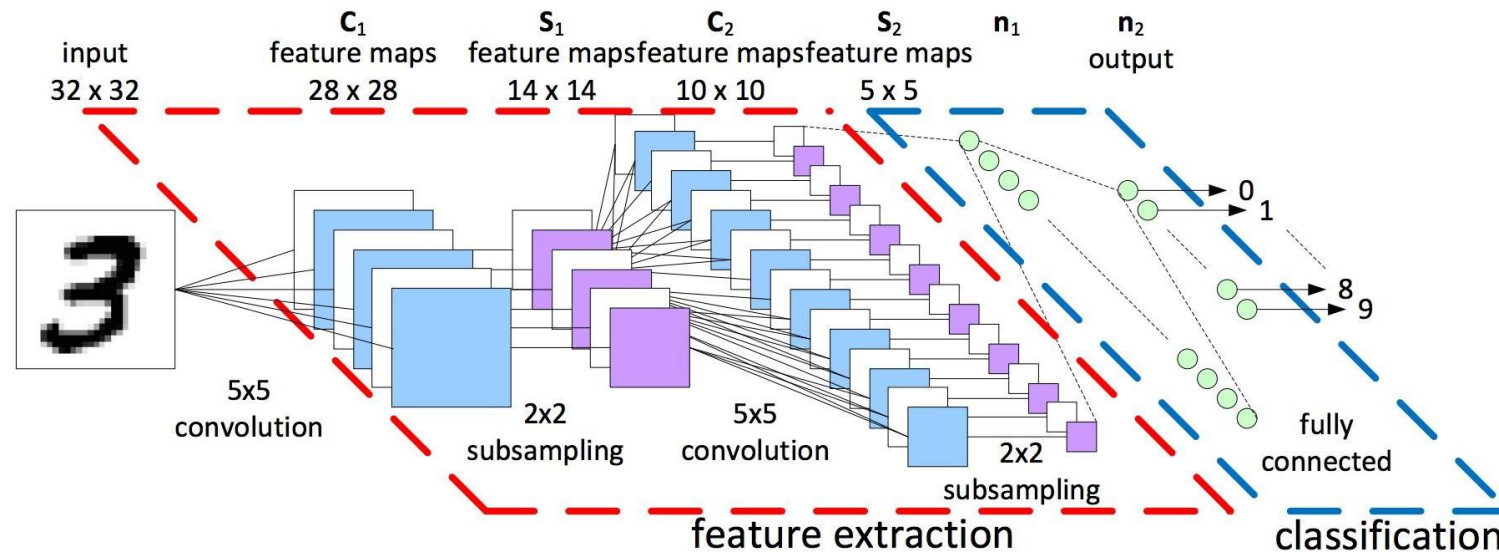


Image Credit: [Deep Learning Methods for Vision | CVPR 2012 Tutorial](#)

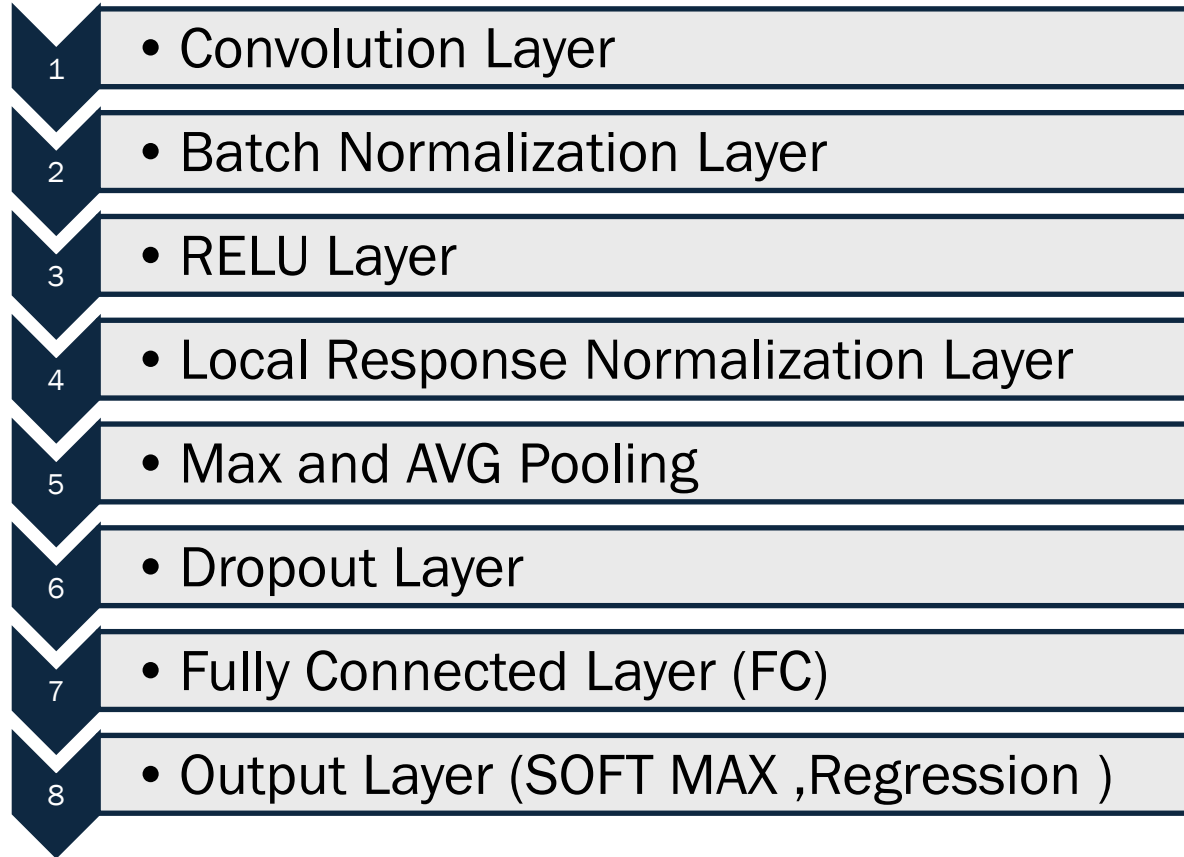
Convolutional Neural Networks

A convolutional neural network (**CNN**, or **ConvNet**) is a class of deep, feed-forward artificial neural networks that explicitly assumes that the inputs are images, which allows us to encode certain properties into the architecture.

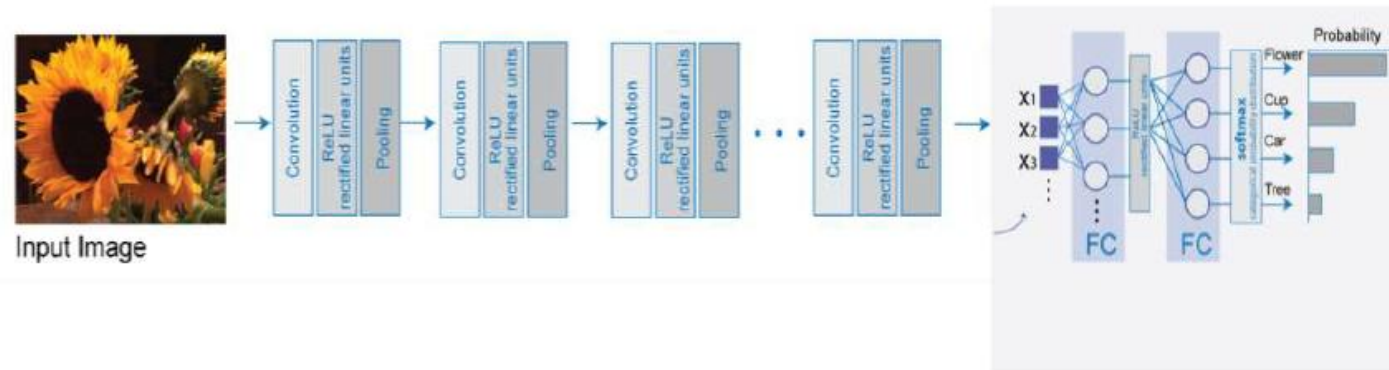


LeNet-5 Architecture (Image Credit: <https://becominghuman.ai>)

CNN Layers



Convolution Neural Networks(CNN)



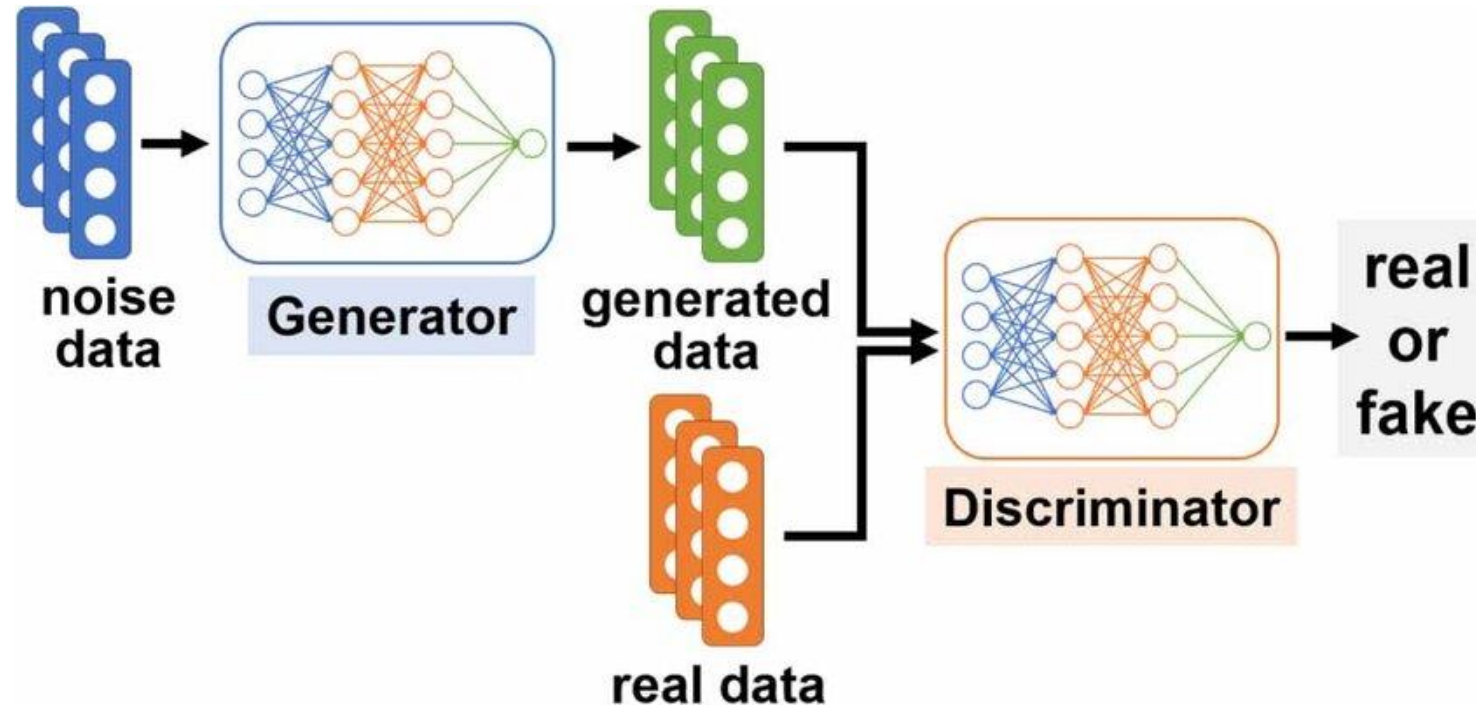
DEFENITION

- Convolutional neural networks learn a complex representation of visual data using vast amounts of data .they are **inspired by human visual system** and learn **multiple layers of transformations** , which are applied on top of each other to extract progressively more **sophisticated representation of the input** .

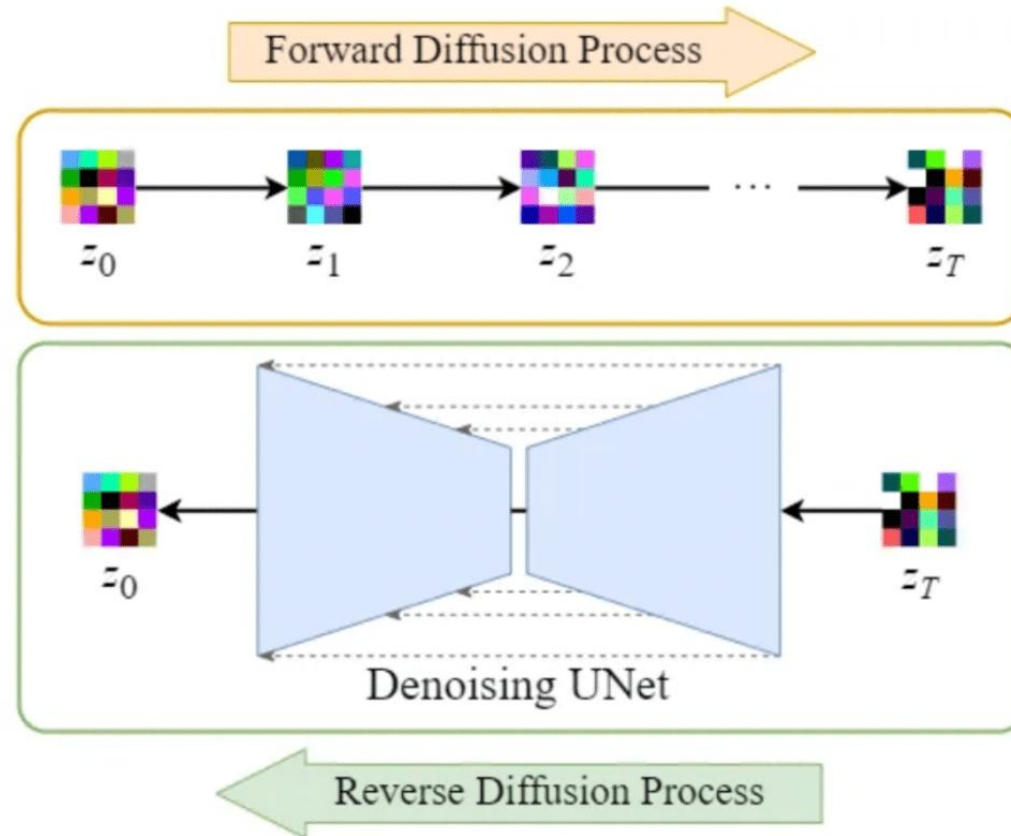
Notes

- Inspired by the visual cortex and Pioneered by Yann Lecun (NYU).
- CNN have multiple types of layers ,the first of which is the Convolutional layer.

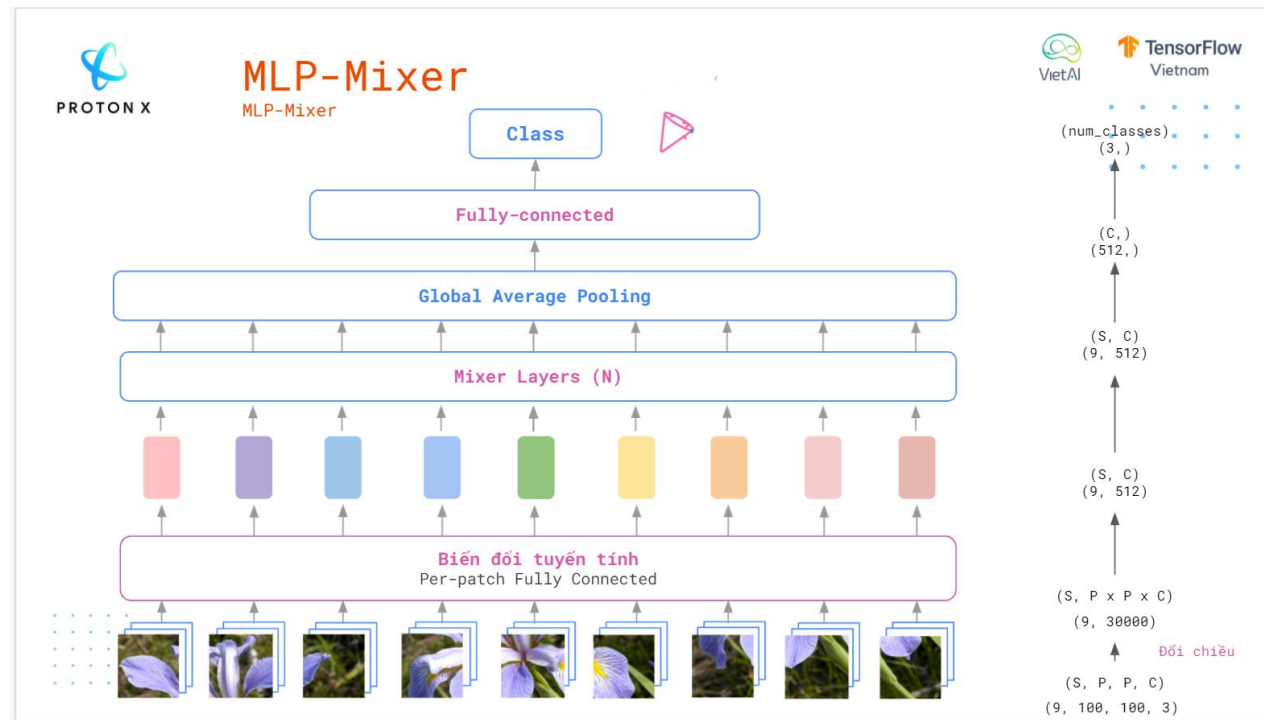
Generative Adversarial Network



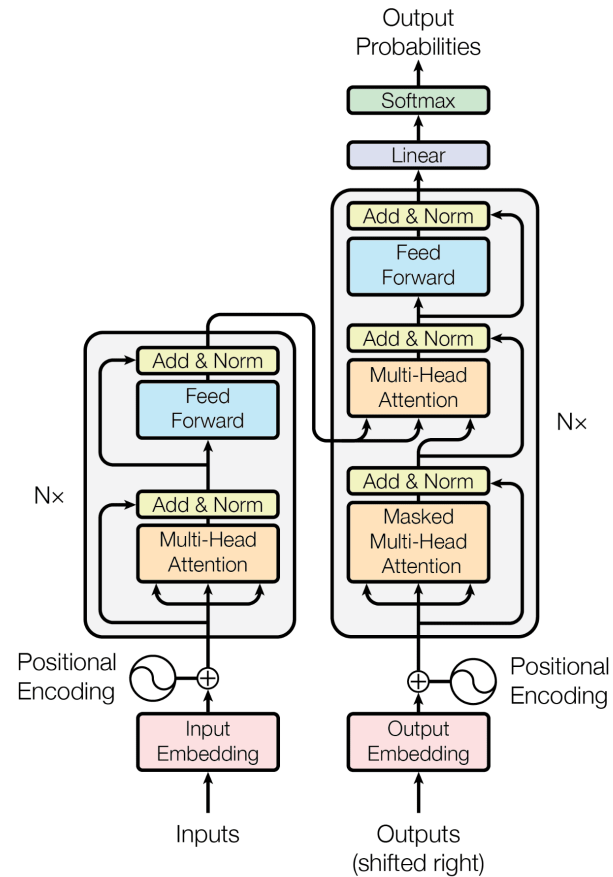
Diffusion model



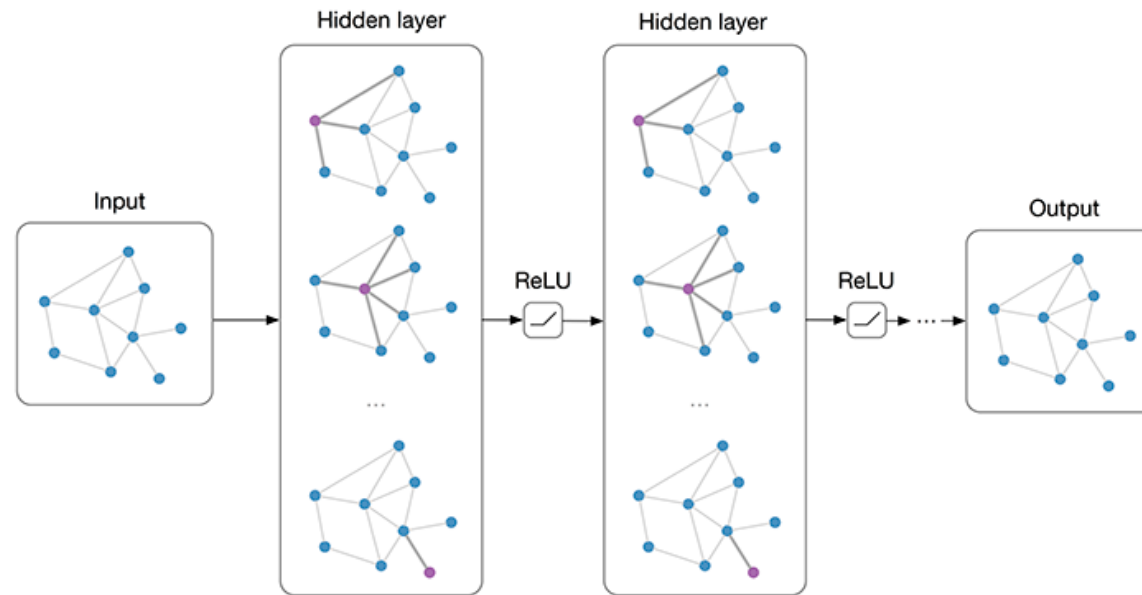
MLP Mixture Model



Transformer



Graph Neural Network



Google Colab

<https://colab.research.google.com/drive/17y5TXNHsz3lj8Y5sbX32q3vbOxYF3L1n?usp=sharing>

Thank You

Reference:

Introduced by : Dr. Amr Rashed Lecturer, Department of Computer Engineering, College of Computers and Information Technology, Taif University

Zhenhua He

